The Instrument Behind the Leaderboard: Re-Execution Noise and Label Reliability in the AIMO Interpretability Challenge

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Abstract

We treat the AIMO Interpretability benchmark, which asks whether a language model's answer to a math problem survives perturbations of the problem, as a measurement instrument, and report three negative results. First, on DeepSeek-R1-0528-Qwen3-8B, inside the official runtime image and pipeline and on the evaluation workers' GPU model, 24 of 40 items change prediction across five executions on the same inputs, and twenty-three executions fall into fourteen numerical states. Second, labels we regenerated under the rule the public labels follow, with our own prompts, decoding and answer key, agree with themselves at Cohen's kappa 0.18 to 0.44 across four models and with the official DeepSeek labels at 0.03; run on our labels, a simulation of the organizers' process overstates this test-retest agreement by about 0.4, so the reliability of the official label remains unmeasured. Third, of thirty-one hypotheses recorded in dated files and tested mostly on DeepSeek, twenty-one died by their kill criterion, and the two probes that survived are largely redundant with problem length. We close with seven recommendations and list fifty-four corrections to our own statements.

1 Introduction

A leaderboard is a measurement: its score is read off an instrument made of a label, an evaluation set, an inference pipeline and the hardware it runs on. When the quantity to be predicted is itself produced by a stochastic process, the instrument can fail in ways that no single submission can see.

We study the instrument of the AIMO Interpretability Challenge, in which a submission predicts, for a given model and math problem, whether the model's answer is robust to perturbations of the problem [1]. We approached the task as a falsification campaign: every hypothesis was recorded in a dated file, and every kill is reported. Of thirty-one entries, twenty-seven carried a kill criterion fixed before they ran[1]: twenty-one died by it, five survived it and one is not established.

24 of 40	0.18 to 0.44	0.03	21 of 27
items whose prediction changes across five identical executions (DeepSeek, default configuration; Section 4)	test-retest kappa of our regenerated labels on four models (Section 5)	kappa between our regenerated DeepSeek labels and the official ones (Section 5)	entries with a kill criterion that died by it; the surviving probes are largely redundant with length (Section 3)

Our findings concern the instrument: re-execution noise with structure (Section 4), a label that our regeneration does not reproduce and whose reliability is unmeasured (Section 5), and apparent

[1] Three of them in a written plan rather than a frozen pre-registration; for H2 the criterion was written after its concordance had been observed (Appendix A).

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signals that trace back, wholly or mostly, to something other than the model's internals, or are largely redundant with problem length (Section 3). Nondeterminism of LLM inference is documented in general [2, 3]; we add its structure and consequences on this benchmark's hardware and pipeline, and two candidate remedies. The organizers reached a compatible conclusion: the best leaderboard scores did not generalize to unseen problems [6]. We make no claim about any participant's method. Appendix J gives the long version of Sections 2 to 7, and Appendix C our fifty-four corrections.

2 The task, the labels and the public data

Task	A submission receives a model identifier, a reasoning effort and a list of problems, and returns one robustness verdict per problem, offline on one GPU within one hour [4, 5]; 70 minutes since 15 October, when evaluation moved to one call per problem [6].
Models	Qwen3.5-4B, Skywork-OR1-Math-7B, Olmo-3-7B-Think and DeepSeek-R1-0528-Qwen3-8B in both tracks; gpt-oss-120b in the Main track only.
Evaluation	Since 16 September, private validation tasks of 21 items (Main) and 14 (Small), on which the binomial standard error of an accuracy near one half is 10.9 and 13.4 points. The final test uses 30 to 50 new problems [5, 6].
Label	Its exact criterion is not published [4]. On all 379 labeled rows of the earlier public datasets it equals the conjunction "no perturbation lowered accuracy" (a majority rule fits them equally well). The main-phase train set thresholds the largest accuracy drop across perturbation families: robust at most 0.10, non-robust at least 0.25 [5].
Public data	Six datasets aggregate to 335 (model, problem) pairs over 174 problems; 52 pairs repeat the DeepSeek measurements under a second model name. DeepSeek is the only evaluated model with substantial ground truth under the earlier rule: 164 labeled pairs, 26 of them derived by us under the conjunction (a majority rule would call 21 of these 26 robust rather than 4).
Two pitfalls	In the three aggregated datasets, the field `permutation_type` is a JSON list if and only if the item is robust, an artifact of the aggregation that is absent at inference. And 20 of the 80 robust published pairs have base accuracy 0 and are robust only trivially under the conjunction, while the public validation sample, `val-sample`, has no labeled pair with base accuracy below 0.05.

3 The falsification campaign

Table 1 summarizes the campaign; Appendix A gives all thirty-one entries with their baselines, numbers and caveats.

Three traps. Rule T gains six points on the public pairs and nothing on `val-sample`, where the trivially robust subpopulation is absent. The oracle H13 chose each target's threshold on the very items it was scored on; forced to rank them, it gave -2.8. And H2, the only warmup hypothesis about a property of the problem that had survived, owed its survival to one measurement published under two model names and counted twice.

Length and probes. The character count alone orders the DeepSeek labels at AUROC 0.705. Our logistic probe on DeepSeek's last-token hidden states reaches 0.710 out of fold and survives its permutation null[2], but it correlates with length at Spearman +0.55: over 20 second-level splits, character count alone gives 0.696, the probe 0.697 and both together 0.714. The organizers' published DeepSeek probe artifact orders our labels at 0.709 under its own aggregation, but at 0.57 once residualized on log character count, and its controls trained on shuffled labels give 0.68. These length controls and artifact analyses were post hoc. Under its own aggregation the artifact thus orders the labels mostly through length; our probe is largely redundant with length, and what it reads is not established. Apart from rule T, no channel we scored in accuracy beats the majority constant by more than about one standard error on public ground truth, mostly DeepSeek's.

[2]With 60 permutations, where 500 were pre-registered; layer and regularization were chosen by inner cross-validation.

Table 1: The campaign in brief. Ground truth is DeepSeek's and the public pairs' unless stated; accuracy differences are in points against each pre-registration's baseline. H1, H7 and T count the 52 duplicated DeepSeek pairs twice; counted once, no verdict changes. Table J.1 gives every result.

What we tested	Entries	Outcome
Text features; robustness as a property of the problem; labels, base accuracy and difficulty order across models	H1 to H5, H7	dead; H2 gained only when the duplicated pairs were counted twice
Official uncertainty features, logistic regression	H10	reaches the proposal's accuracy, within its 0.692 ± 0.101 (method substituted): 0.7744 against a majority constant of 0.756
Features cached for other models; per-target thresholds	H12, H13	H12 dead; the H13 ceiling withdrawn, -2.8 points when forced to rank
Our perturbation families, translation, self-consistency, the organizers' families regenerated (pilot), representation shift	H19, H22, H30 to H33	dead: kappa at most 0.02, AUROC at most 0.615
Probes on DeepSeek's hidden states: ours and the organizers' published artifact	H36, H35	survive, largely redundant with length
Trivial-robust rule, post hoc	T	a trap: +6.0 points on the public pairs, +0.0 on `val-sample`
Identical bundles re-executed (warmup, 17 items)	H18	noise confirmed: 11, 9, 11 and 11 correct

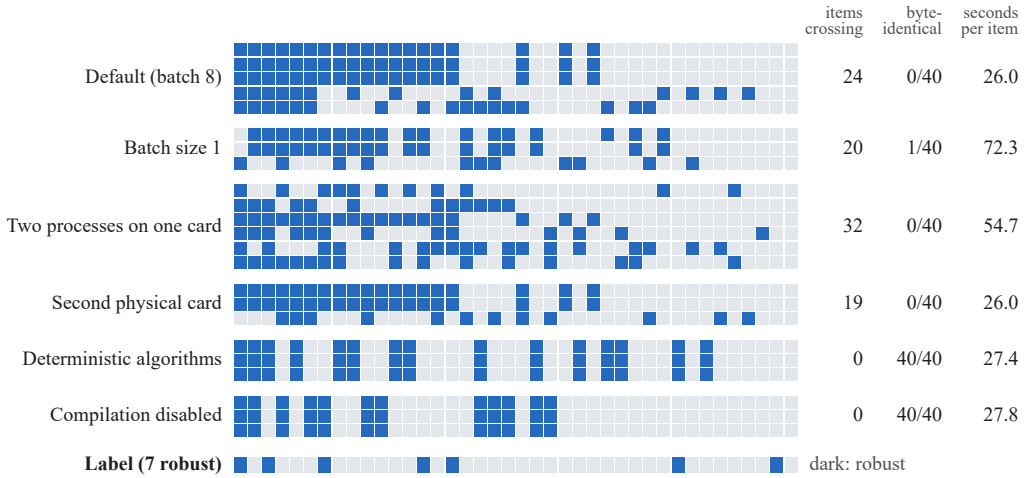


Figure 1: Predictions of the official pipeline for forty public problems (columns) in twenty-three executions on DeepSeek (rows), by configuration; the last row is the label, dark cells are robust, and columns are sorted by how often they are predicted robust. At right: items whose prediction changes across the configuration's runs (more runs, more crossings), items whose generated text is byte-identical in all of them, and seconds per item. Table J.2 gives the mean number of crossings per pair of runs.

4 Re-execution noise on the evaluation workers' GPU model

On a virtual machine lent by the organizers, with two NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs (the evaluation workers' model), we ran forty public problems through the official runtime image, extraction and regressor, unmodified, on DeepSeek (Figure 1).

Fourteen states. The twenty-three executions fall into fourteen numerical states, byte-identical within a state. The state belongs to the process, not to the card: two runs launched in the same second on one card landed in different states, while two of three runs on the second card reproduced the first card's default state exactly. We infer, without proof, that the state is fixed when the process starts;

Table 2: Agreement of regenerated labels under the conjunction: Cohen's kappa, with 95% percentile bootstrap intervals over problems. Robust rates are over all 164 problems (the 4 without accepted variants count as not robust); kappas over the problems labeled in both series (159 for A and B, 160 for A′ and the official labels). Among these models, official labels under the earlier rule exist for DeepSeek only; Table J.3 adds the main-phase rule.

Model	Robust in A	A and B	A and A′ (new seed)	Official labels
Skywork	72.6%	0.439 [0.28, 0.59]	0.789 [0.67, 0.89]	none
DeepSeek	37.8%	0.413 [0.26, 0.55]	0.464 [0.33, 0.59]*	0.025 [−0.12, 0.18]
Olmo	65.9%	0.378 [0.21, 0.54]	0.653 [0.52, 0.78]	none
Qwen3.5-4B	9.1%	0.178 [−0.06, 0.39]	not run	none

*On DeepSeek, A′ also received the answer-format line of Section 5.3, so this entry mixes seed and prompt.

two pre-registered predictions about the states failed (Appendix J). As only 7 of the 40 problems are robust and crossings largely cancel, accuracy moved much less than predictions (21 to 24 correct under the default configuration on either card, 24 to 28 with two processes on one card), always below the 33 of the majority constant.

Candidate remedies. In three sequential runs on one card, either deterministic algorithms (PyTorch's, in warn-only mode, with deterministic cuDNN and a fixed cuBLAS workspace; flags in Table J.2) or disabling compilation made generation byte-identical at 5 to 7% runtime cost. Neither was tested under contention or across cards, and the default configuration also gave three identical sequential runs before its fourth diverged: these are candidates, not fixes. Nor does determinism buy correctness: the two configurations, run on different cards, settled on different states, predicting 15 and 12 of 40 problems robust and agreeing on only 21.

The official score channel. In an exploratory measurement, not pre-registered, we extracted the official regressor score three times for the 174 public problems with the same code and image. One of the three runs set the cuBLAS workspace variable; the other two correlate at Spearman 0.24 over the 164 labeled problems and give AUROC 0.487 and 0.602, a difference that is not significant: the ordering does not reproduce, which is not a demonstrated change of signal. As 137 of the 164 problems are in the regressor's training set, this concerns the stability of its inputs, not new problems.

5 Can the label be reproduced?

We identified the organizers' perturbation families in the public data (rephrase, rename, domain, distract, typos), regenerated them with gpt-oss-120b prompted with the public examples, and gated each variant with gpt-oss-120b[3]. An answer counts as correct when it matches gpt-oss-120b's answer on the original problem, as the public data carry no answers. On the organizers' machine, with vLLM 0.28.0 outside the official image, we labeled four models with ten variants per family in two independent draws (A and B; about 40 accepted variants per problem), ten samples on the base problem and one per variant, and re-sampled draw A's variants with a new seed (A′).

5.1 More variants do not help, and our noise does not explain the gap. Subsampling DeepSeek's variants from 2 to 40 per problem, the mean kappa(A, B) stays between 0.36 and 0.47 with no upward trend, and the mean concordance with the official label between −0.02 and +0.03. Corrected for our own noise, 0.025 becomes 0.039 (about 0.27 at the top of its 95% interval): our noise could hide at most a weak agreement. Under the main-phase rule, pre-registered on the same answers, test-retest stays below 0.60 on every model.

5.2 What drives the instability. On Skywork, re-sampling the same variants with a new seed agrees far more than drawing new variants (0.79 against 0.44), so there the label depends mainly on which variants are drawn. A plug-in simulation of the organizers' process from their published prediction counts gives a test-retest kappa of 0.94, but it resamples answers to the same variants and never lets a robust item fail; on our draw A it gives 0.64 to 0.84 against a measured 0.18 to 0.44, overstating by

[3]The text must differ, the length ratio lie between 0.7 and 4.0 and the solver's answer not change; prompts in Appendix I.

about 0.4 on every model. The reliability of the official label therefore remains unmeasured. The official DeepSeek labels also come from a different sampling regime [5] and decoding [4], so the data show something narrower than a verdict on the label: variants produced by our process measure something else. The organizers have decided not to publish their labeling code [6].

5.3 An unspecified prompt moves the label. Fed the bare problem text, Qwen3.5-4B closes its answer in the expected format in only 5 of its 1,640 base samples in draw A; with one instruction line requesting the format, the share of problems passing the label's base condition[4] moved from 0 to 145 of 164 (88%). A label defined by correctness is not comparable across prompts that the benchmark leaves unspecified.

6 Our entries, and the evaluations the organizers requested

V8, the entry that does nothing. Pre-registered on 13 September, its entire logic is `return [len(p) >= 98 for p in problems]`, with the threshold at the lower median of the 174 public problems and the direction fixed on the public pairs. On the original main-phase tasks it scored 11/21 and 10/19, identical to our measuring entry v6 (the official extraction and regressor on each model's own greedy generation) and to the organizers' own entries on the leaderboard; none of these entries scored above the majority constant. On 21 and 19 items these ties are compatible with length carrying little signal (95% Wilson intervals 0.32 to 0.72 and 0.32 to 0.73) without showing that it carries none. A one-line rule that ties a pipeline loading up to five models for 8.5 to 14 minutes is a measurement of the pipeline.

The evaluations requested for reports [5]. On the public main-phase train set (revision of 17 September), V8 scores 46/71, below the 52/71 of the all-robust constant. On the replacement tasks of 16 September it predicted robust for every item: 10/21 and 7/14, the base rates. On 28 October, after the change of 15 October [6], we resubmitted V8 unchanged (Codabench 1016713, Main; 1016730, Small) with an all-robust constant as control (1016732, 1016735); both predicted robust for every item and scored 10/21 and 7/14. For the final test we nominated V8 (1016713) and the constant (1016732), both on the Main track; their scores were not yet available when we submitted this report. Our pre-registered predictions will be scored there (Appendix B): P1, that the top-ranked entry's final accuracy will lie within two binomial standard errors of the test set's majority constant, and P2; if either fails, the label has learnable structure that we did not find.

7 Recommendations

- **R1. Fix the numerical state, and test the fix under load.** Verify either remedy of Section 4 under the workers' real load and across cards, and document the chosen setting, which selects the fixed point; isolating submissions on a card may matter as much.
- **R2. Repeat only where it decides something.** At the warmup's re-execution standard deviation (1.0 item on 17, from four runs, itself uncertain), a one-item gap needs about eight runs: repeat only entries near a prize boundary.
- **R3. Publish the labeling prompt,** at least after the close.
- **R4. Measure the label's reliability directly,** with two independent draws of variants labeled by the official process.
- **R5. Show the floors and the noise on the leaderboard:** a row per constant predictor, a re-execution standard deviation and binomial intervals.
- **R6. Require a length control in white-box ablations:** without it, removing the white-box component cannot separate its signal from length.
- **R7. Remove the leaking field** `permutation_type` from the aggregated public datasets.

Code and data. Our code, bundles, pre-registrations, outcome files and full report (v2.4) are released at https://github.com/Dan23RR/aimo-instrument; Appendix K gives release terms, bundle hashes and our Codabench submissions.

[4] At least 6 of 10 samples agreeing with the reference answer.

Limitations

Ground truth under the earlier rule exists for two evaluated models, and every probe and every concordance with the official labels in Sections 3 to 5 is measured on DeepSeek, 26 of whose 164 labels we derived; these results are assumed, not shown, for the other models. The main-phase train set, at the revision we used (Appendix H), labels all five models but has 3 to 37 labeled rows per model, and apart from the evaluation of V8 and a descriptive comparison with our regenerated labels on the five labeled rows that fall on our problems and models, we do not use it. Section 4 uses 40 and 174 problems and one model; the fourteen-state structure is established, its cause inferred, and the remedies were tested only in sequential runs, each on one card and the two on different cards, so configuration and card are not separated. The regeneration of Section 5 used one generator, gated by itself, with our own answer key, sampling counts, decoding and prompts, so its negative concordance is a fact about our process, not about the families. Several analyses were first run from scripts that were not saved and were re-created and saved later, and a few figures rest only on a dated record (Appendix D). Nothing here estimates performance on the final held-out problems, beyond the predictions of Appendix B. The measurements of Sections 3 (probes), 4 and 5 were taken on the organizers' virtual machine; on 7 September, and again on 2 November, we asked them whether such measurements may appear in this report, and on 2 November they answered that this is acceptable (Appendix K). The five-page main text condenses Appendix J, which states the caveats of each section more fully.

AI usage statement

In this work, we used one generative AI tool, Claude (Anthropic), through Claude Code, for ideation, coding, analysis and writing. Ideation: it drafted the pre-registrations, including hypotheses, predictions and kill criteria, and searched and summarized the literature and the public records, including the organizers' Discord channels and the author's correspondence with them. Coding: it wrote the solution code of our bundles and the analysis and extraction scripts, ran them, and operated the organizers' virtual machine over SSH. Analysis: it computed and interpreted the results and ran automated multi-agent adversarial reviews of hypotheses, drafts and code, which found most of the corrections in Appendix C. Writing: it drafted the correspondence and this report, including its condensation to five pages. It also made most of our Codabench uploads through the author's browser at the author's instruction, with two exceptions: one resubmission (925822, 14 September), which it made without a specific instruction, and one submission that the author had not approved (1016734, 28 October): uploading the constant's Small bundle, it sent it to both tasks, probably because the form had re-selected them after its check. Language models were also components of our method: gpt-oss-120b generated and gated the perturbation variants and supplied the reference answers of Section 5, and phi-4 translated problems for H22 (Appendices H and I). We did not use generative AI tools to make the final decisions on the direction of the work, on the submissions (with the two exceptions above), on the nominations for the final test and on the claims of this report, which were the author's, and the author sent all correspondence personally. The author has reviewed all AI-assisted work and takes full responsibility for the final content of this report, including AI-assisted text, claims, code and other artifacts.

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References

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[5] aimo-interp/getting-started and aimo-interp/baselines, GitHub repositories: README and issues #6 and #10 of getting-started; aimo-interp datasets on Hugging Face, including `train-main-v2` and `val-sample`.

[6] AIMO Interpretability Challenge Discord, #announcements and #qa: 5, 15 and 22 September, 9, 15, 16 and 20 October 2026.

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A The hypothesis table

Carried over verbatim from the full report (draft v2.4, 23 September 2026), which is released with this report. Section and appendix numbers in this appendix refer to that report; Appendix E maps them to this one.

Validation that fits on some problems and tests on others (H1, H10, H11b, H20, H23, H36) is grouped by problem: no problem appears on both sides of a split. The cross-model tests hold out the model but not the problem. In H7, 234 of the 335 test rows have their problem in training, 198 of them with the same label and base accuracy, including all 52 DeepSeek rows, which are exact copies of training rows published as `qwen3-8b:low`, so part of that test is in-sample (with them counted once, H7 is still +0.0). In H12 to H15, 70 of the 109 test pairs have their problem in training, where the classifier is trained on the same DeepSeek feature vectors labeled by other models, an overlap whose effect was not measured and need not favour the hypotheses, since another model's label on the same problem did worse than the per-model majority in H3. H7, H12, H14 and H15 died anyway, and H13 is withdrawn. Warmup rows are as in v1.0 except where later drafts corrected them (Appendix D, items 6, 21, 24 to 29, 51 and 54). Draft v2.2 made H2 die once the duplicated rows are counted once, moved H13 from dead to a withdrawn oracle ceiling, qualified H16 as within re-execution noise, moved H17 from survives to not established, marked H6 and T as having no kill criterion and corrected the descriptions of H3, H17 and T; this draft describes the classifier actually run for H10, corrects the description of H14, marks H18 as having no kill criterion, adds to H2 the result of E1's original test and to H13 its value without constant predictions, gives H1, H7 and T with the duplicated rows counted once, gives H7 against the baseline its pre-registration named, and describes the rows on which H17's threshold was fitted. Main-phase hypotheses are new.

ID	Hypothesis	Result versus its baseline	Verdict
H1	Problem text predicts robustness (32 features, grouped 5-fold)	0.675 vs 0.761, −8.7 (0.682 vs 0.756, −7.4, with the duplicated rows counted once)	dead
H2	Robustness is a property of the problem (cross-model concordance)	z = 0.88 and −1.3 points against the per-model majority (bar 4.0) once the 52 rows published under both `qwen3-8b:low` and the DeepSeek name, identical in label and base accuracy, are counted once, and z = 1.81 under E1's original global-rate test (bar 2); z = 4.04 and +3.4 only when they are counted twice	dead
H3	Model A's label predicts model B (56 ordered model pairs over 10 problems, 560 predictions)	−7.3	dead
H4	A's base accuracy predicts B's robustness	−11.8	dead
H5	Difficulty order transfers (half-batch rule)	−25.4	dead
H6	Same, oracle k	−13.0	ceiling, negative (no kill criterion)
H7	Base accuracy beats the prior on held-out models	+0.0 against the global prior of the training models, used where PREREG_E4 named the per-model majority (−1.2 against that; +0.0 and −1.4 with the duplicated rows counted once)	dead
T	Trivial-robust rule (robust iff base acc < 0.05), formulated after the data of H9	+6.0 over the 335 published pairs (+7.1 with the duplicated rows counted once), +0.0 on `val-sample`	trap (no kill criterion)
H10	A classifier on the official uncertainty features reaches the accuracy the organizers' proposal reports for their baseline (grouped CV)	0.7744 against a majority constant of 0.756, inside the 0.692 ± 0.101 that the proposal reports (best of several classifiers, stratified 10-fold CV on its sample validation set); run as a logistic regression on 13 of the 14 features, three of them constant in our extraction, where PREREG_E6 specified the official regressor and threshold	confirmed (method substituted; not declared at the time)
H11b	Length, saturation and trend features add signal	−1.8 / −2.5	dead
H12	Cached model's real uncertainty features predict other models	−16.5	dead

ID	Hypothesis	Result versus its baseline	Verdict
H13	Same, with a threshold chosen per target on its own items	+7.3 (+5.5 when constant predictions are excluded), all from thresholds chosen on the items they are scored on; −2.8 once the number of predicted positives is fixed at the true count (E8)	ceiling, withdrawn (no kill criterion)
H14, H15	Calibration by the target's public robust rate (H14) or by the global rate (H15)	−2.8, −10.1; each target's public rate is its rate on the tested items, so H14 coincides with the oracle-k ceiling of E8	dead
H16	Self-measurement of every model beats the constant	10/17 > 9/17 on one execution, one item, inside the spread of H18	survives its criterion, within noise
H17	A gpt-oss threshold calibrated on the ten gpt-oss rows of `val-sample`, one of them robust (6.2), other models unchanged, beats H16	11/17 > 10/17 on its first execution; three re-executions of the same bundle gave 9, 11 and 11, and 9/17 meets the kill criterion	not established
H18	Identical bundles score differently across executions	{11, 9, 11, 11}	confirmed (no kill criterion: PREREG_V7 fixes the measurement, not a criterion)
H19	Our five mechanical perturbation families reproduce the official label (DeepSeek, 99 pairs)	accuracy 0.354 vs 0.788, kappa **0.02**	dead
H20	A threshold tuned on our label transfers to theirs	0.212 vs 0.788	dead
H22	Translation-stability (it/fr/de) reproduces the official label (77 pairs)	0.507 vs 0.740, kappa −**0.10**	dead
H23	A threshold tuned on translation-stability transfers	+0.0	dead
H29	Representation-distance features are repeatable across batch size and padding (gate; E11 plan, not a frozen pre-registration)	ICC(2,1) 0.997 to 0.999, from the dated outcome file of E11 (the per-item output was written to the Modal volume and is not in the local record)	passes
H30	Residualized representation shift under perturbation predicts the label (E11 plan)	AUROC 0.615: below the death threshold of 0.655 fixed in the plan (the null p95 over its planned cells), above 0.604, the uncorrected p95 for a single statistic (permutation p = 0.030); n_char 0.705	dead
H31	Directional coherence of the shift predicts the label (E11 plan)	0.555 (permutation p = 0.29)	dead
H33	Self-consistency over 8 samples predicts the label (164 pairs)	AUROC **0.502**	dead
H32	The organizers' families, regenerated, reproduce their label (157 pairs; four of five families, `domain` excluded by the pre-registered 60% acceptance gate)	kappa −**0.001** (−0.030 with all five)	dead
H35	The organizers' published probe artifact orders our ground truth	0.709 with its own inference aggregation, mostly problem length (0.57 once residualized on log character count); 0.739 with the pre-registered one (0.62 once residualized, as for our probe); its control probes, trained on shuffled labels, give 0.68 by themselves at layer 31 (8.3)	survives, largely explained
H36	Our linear probe, nested CV, orders our ground truth	0.710 OOF against a reduced nested null (p95 0.574 with 60 permutations and two of four regularization values, where 500 and the full grid were pre-registered; 0.570 with the full grid when re-run on 23 September); about +0.02 over character count in a second-level cross-validation (8.3)	survives, largely explained

ID	Hypothesis	Result versus its baseline	Verdict
E13b-1	Under the main-phase rule our regenerated label is test-retest reliable (kappa(A, B) $\geq$ 0.60 on at least 3 of 4 models)	0.34 to 0.56, none reaches 0.60 (7.5)	dead
E13b-2	Under the main-phase rule it agrees with the official label (DeepSeek, kappa $\geq$ 0.40)	0.02 [$-$0.09, 0.13] (7.5)	dead
E13b-3	The labeling rule drives the instability (kappa(A, B) gain $\geq$ 0.15 on at least 3 of 4 models)	gain $\geq$ 0.15 on 2 of 4, and on 0 of 4 when both rules are read on the same problems (7.5)	dead

Of the thirty-one entries in the table (thirty rows; H14 and H15 share one), twenty-seven carried a kill criterion written before they ran: twenty-one died by it, five survived it (H10, H16, H29, H35, H36) and one (H17) is not established. Three of the twenty-seven (H29 to H31) had their criterion in the written plan of E11, fixed before the runs, rather than in a frozen pre-registration. H2 is weaker than the rest: its concordance (56 of 73) had already been observed when PREREG_E1 was written, and the per-model null and the bar used in its row were added in E3, after the first run; with the duplicates counted once, H2 also fails E1's original test (z = 1.81 against 2). The other four entries have no kill criterion of their own: H6 and H13 are oracle ceilings (H6 came out negative; H13 came out +7.3 and was withdrawn in E8), T is a rule formulated after seeing the data of H9, stopped by the check on `val-sample` announced in E4, and H18 is a repeat-run measurement whose pre-registration fixes what to measure but no criterion.

Executed but not in the table: H8, the E4 oracle ceiling (0.776, +1.5 points against the global constant and +0.3 against the per-model majority; +1.8 and +0.4 with the duplicated rows counted once); H9, whose outcome became rule T; and the V4 allocation entry (909843, Appendix B), whose pre-registration stated that a tie at 9/17 was impossible unless the model path on the DeepSeek items failed. It tied at 9/17. None of the three explanations recorded then was verified; the first, drift on the DeepSeek items, is the kind of re-execution noise later measured in H18. H1, H7, H8 and T were computed on all 335 published pairs, so the 52 duplicated DeepSeek rows count twice; counted once (283 pairs), no verdict changes, and the values are given above. Predictions attached to measurement experiments are reported where they belong: against our written predictions, batch size 1 does not stabilize generation and the configuration without compilation lands in a state of its own (8.1). E15, the experiment behind H35 and H36, also carried a written prediction, that both probes would die; both survived their null (8.3).

E11 departed from its plan in ways not declared at the time. The plan listed three further primary tests, noise injection (H32), a geometric control (H33) and a trajectory-angle control (H34), which were never run; distributional stability on the unlabeled models (H35) was conditional on a channel passing and was correctly not run. The plan required Holm over five primary tests at alpha 0.010; the analysis script announces Holm over the two tests it ran but computes no correction, and H30, with a permutation p of 0.030, would not have passed Holm over those two even at the alpha of 0.05 implied by the script's p95 threshold (0.030 against 0.025). The analysis used all 164 problems, where the plan held out 40 as a confirmation set and ran the primary analysis on 124. It computed each channel's AUROC directly on the scalar, with the orientation chosen on the data, rather than out of fold from an L2 logistic regression with C = 0.05 over 5 folds and 20 repetitions. It replaced the six-feature text baseline of H28 with the AUROC of character count alone. It omitted the two analyses that the plan (its section 4.1) made obligatory next to every reported AUROC, the same AUROC stratified on quartiles of the number of perturbed predictions and the reference AUROC of the bookkeeping counters (0.862 out of fold, not available at inference). And it extracted only the primary depth (0.80) of the three the plan fixed. On the 164 problems analyzed, H30 is below the plan's threshold of 0.655 and trails character count by 0.09, and H31 does not reach its null. Identifiers H32, H33 and H35 were reused in E12 and E15 for the hypotheses listed here; H34 was not reused. H21 and H24, the second stages of E9 and E10, were to run only if H19 and H20, and H22 and H23, survived, and were correctly not run. The plan's H25 to H28 are not in the table: H25 restated as a confirmation the A100 anomaly that 8.2 still reports (AUROC 0.36 in the rule's orientation), traced to the feature that our own extraction returned as 1.0 on every item (2.3b), and its prospective part, a simulation with a kill criterion, has no recorded outcome; H26 (the aggregation defect of 2.3a) and H27 (a label-noise ceiling) were measured before the plan was written; and H28,

a text-only baseline, was also measured before the plan (character count 0.687 out of fold, as recorded there), while its planned non-inferiority test against the official features was never run.

Two of the three methodological notes of v1.0 stand: in-sample signal refused to generalize twice, and there is the distribution trap of rule T (2.4). The third, that grouped validation is what killed H1, is withdrawn (Appendix D, item 12): under an ungrouped 5-fold split, with the same code and data, the same text model also stays below the majority constant (-0.9, -0.3 and -2.1 points over three seeds, against a bar of 2.7), so grouping deepened the loss but did not cause it. The main phase added a fourth, in H30: a control matched on the nuisance variable and decided before the data (token-matched neutral padding) separated a length signal from a fragility signal; in 8.3 the same separation needed a control we added only after seeing the data, which is why such a control belongs in the pre-registration. The verification of draft v2.1 added a fifth: the public datasets publish one measurement under two model names, and counting it twice produced the only surviving warmup hypothesis about a property of the problem (H2).

Sections 4.1 to 4.3 of draft v1.0 are not reproduced in this draft; their content is in the dated files PREREG_E1 to PREREG_E8 and PREREG_E5_ESITO, which will be released with the final version (Section 13).

B Pre-registered predictions to be scored against the final evaluation

P1 and P2 were written on 7 September 2026 (PREREG_E12), P-V8 on 13 September (PREREG_V8) and the P-V8b predictions on 21 September (PREREG_V8b). They are translated from the dated files and otherwise unchanged, except that P2 there names the participant then leading and quotes those entries' figures, which we do not. P2 asks whether a leaderboard figure survives new problems, not how it was obtained, and no inference about any participant's method will be drawn whatever the outcome.

- **P1.** The top-ranked entry's accuracy on the final held-out problems will lie within two binomial standard errors of that set's majority-class constant.
- **P2.** The final accuracy of the participant whose entries led both tracks on 7 September will be lower than those entries' leaderboard figures of that day by more than two standard errors.
- **P-V8.** If the organizers had neutralized length, the deterministic entry would score between 0.45 and 0.55 on both tracks (stated beforehand as the most likely outcome); if they had not, and the set was balanced, about 0.66; in either case with coverage 1.0, no invalid predictions and a runtime under 60 seconds. Main 0.52 and Small 0.53 fall inside the first band; on 21 and 19 items, however, 0.66 lies inside the 95% intervals of both scores, so the scores do not exclude the second. The clause on coverage, invalid predictions and runtime was not scored at the time: each evaluation on the task the bundle was built for had coverage 1.0 and no invalid predictions, and the bundle logs no runtime.
- **P-V8b**, on the replacement tasks: the old interface works with the new ingestion; the same bundles score 6 to 15 of 21 and 4 to 10 of 14; each finishes in under 60 seconds. Observed: coverage 1.0 and no invalid predictions; 10/21 and 7/14, inside the bands but degenerately, since the rule predicted robust for every item; each evaluation was marked finished within about a minute of upload, queue and scoring included, an upper bound on the runtime.

C Corrections to our earlier statements

Carried over verbatim from the full report (draft v2.4, 23 September 2026), which is released with this report. Section and appendix numbers in this appendix refer to that report; Appendix E maps them to this one.

Full report, Section 11: Corrections since draft v1.0

Appendix D lists fifty-four corrections to statements of our earlier drafts and dated files, each made in place in this draft. Items 1 to 4 were listed in draft v2.0 (14 September), which draft v2.1 overwrote and which is not preserved as a separate file. Items 5 to 18 come from an adversarial verification of draft v2.1 against the dated files and the correspondence (21 September). Items 19 to

53 come from a second adversarial verification, of draft v2.3 (23 September), which checked every factual sentence again against the dated files, the saved outputs and the public records of Codabench, GitHub, Hugging Face and the competition website, recomputed every figure it could, and also extended items 1, 4, 6, 8, 10 to 14, 16 and 17 with corrections of its own. Item 54 comes from a third adversarial verification, of this draft before its release (23 September), which also revised items 1, 4 to 6, 11 to 22, 29 to 32, 34 to 39, 41 to 43, 46 and 49 to 53, adding to several of them corrections of statements of earlier drafts (for example items 18, 31, 42, 43, 49, 51 and 52); its other findings, on sentences of this draft and on the dated notes, were corrected before release. A fourth, narrower check of the changes made after the third found further problems of attribution and wording, which revised items 1, 18, 21, 22, 34, 42, 51, 53 and 54 and the dated notes; they too were corrected before release. The scripts the verifications required were saved on 21 and 23 September (Section 13).

The corrections that change a conclusion include these: the classifier behind H10 was not the official baseline (item 24); the H13 ceiling came from thresholds chosen in-sample, not from a constant (item 25); the E11 analysis applied no multiplicity correction, and H30 would have failed Holm over its two tests (item 30); 26 of the 164 DeepSeek labels used as ground truth were derived by us (item 33); the organizers' DeepSeek labels come from a sampling regime of about ten answers per variant, not one (item 34); the organizers' probe artifact orders our labels, under its own inference aggregation, mostly through problem length (0.57 once residualized on log character count), as its own control probes trained on shuffled labels do (item 46); the plug-in simulation overstates reliability on our own labels by about 0.4 (item 41); on Olmo the seed accounts for about half of the instability under either rule, so which variants are drawn does not dominate there (items 39 and 43); the re-execution drift makes the official score channel's ordering irreproducible between identical runs but is not shown to change whether the channel carries signal (item 45); the cause of the numerical states is not identified (item 44); the budget guard is not shown to prevent timeouts (item 23); and the gap to the official DeepSeek labels is not only one of severity (item 43).

Our claim now is narrower and measured: apart from rule T, which exploits a subpopulation absent from `val-sample`, none of the channels available to us that we scored in accuracy, and not the organizers' own probe artifact, clears the majority constant by more than about one standard error on public ground truth. We make no claim about methods we have not seen. Apart from the public leaderboard figures cited in Sections 3.3 and 6, in the introduction of Appendix C and in Appendix D (items 1, 17 and 52), and predictions P1 and P2 in Appendix C, all of which concern scores and submissions, not methods, nothing in this report should be read as a statement about any participant.

Full report, Appendix D: Corrections to our earlier statements

Each item names the section corrected and the drafts or dated files in which we found the statement; items 5 to 18 refer to draft v2.1 unless they name another. The correction is made in place in the section cited, and the dated files carry correction notes of 21 and 23 September.

1. Drafts v1.0 to v2.3 (Section 3.3) explained a flat leaderboard by the shipped baseline being a constant on unloadable models (v1.0: "explains why the mid-warmup leaderboard clustered at exactly 0.53 and 0.47"; v2.1 to v2.3: "which is why the mid-warmup leaderboard clustered at the two constant values"); our message of 30 August had described that constant behaviour on the warmup set only, without the leaderboard inference. The baseline falls back for lack of a shipped artifact, not of a cached model, and it scored 9/17 itself, so it could not account for the rows at 0.47; an explanation of other participants' scores by a presumed method is withdrawn (3.3). In the main phase the leaderboard moved: by 13 September individual submissions had reached 0.95 on Main and 1.00 on Small (recorded in PREREG_V8). On 16 September both evaluation tasks were replaced (21 and 14 items). Until the morning of 21 September, before our uploads, no submission on the new tasks had exceeded 14/21 on Main or 9/14 on Small, against 10/21 and 7/14 for the all-robust constant (6.3); at 10:02 UTC that day a submission reached 11/14 on Small, and on 23 September at 03:40 UTC one reached 18/21 on Main. Under the declared rule the public table lists every scored submission as its own row (614 rows on 14 September, fifty per page, as recorded in PREREG_V8), so the printed rank is per upload, not per participant. The narrower claim that replaces the explanation is stated in Section 11, with the exception of rule T that drafts v2.1 to v2.3 omitted there.

2. During 8.1, after three identical repetitions, we concluded in an internal message that the drift came not from the GPU or the pipeline but from the shared environment. Repetitions four and five falsified that within the hour; the conclusion never left the project. The corrected finding is in 8.1.

3. An internal analysis had stated that the shipped decision threshold "costs 43 points" because it predicted 66.5% robust on our A100-extracted features. On the competition hardware it predicts 25.6% robust in two of three identical extractions and 33.5% in the third, against a 24.4% rate, and costs 12 to 16 points against the constant, not 43. The 66.5% came from our own re-implementation of the extraction (item 5); the claim was never published and is withdrawn here so that the record is complete.

4. In an internal note of 13 September (PREREG_V8, withdrawn there on 14 September) and in two paragraphs of PREREG_E13 we inferred from the 0.41 test-retest of our regeneration that no method could exceed roughly 0.7 accuracy, and that leaderboard scores above it were therefore anomalous. The inference used our process's reliability where the organizers' was required, and the arithmetic was also wrong: with a reliability of 0.41 the bound on agreement is its square root, 0.64, which allows 0.82 accuracy on a balanced set, not 0.70. The inference is withdrawn.

5. Draft v2.1 (Section 2.3(b)) reported that `generation_frac_selected_is_top1` is identically 1.0 at inference, so that one fifth of the forest routes every production sample the same way (the feature carries 17.1% of the splits). That was true only of our own A100 re-implementation of the extraction; in the official extraction on the organizers' hardware the feature is never 1.0 and falls mostly inside the split range. The defect is withdrawn, with the half of R5 that depended on it.

6. H2, the only warmup hypothesis about a property of the problem that had survived, survived only because the same 52 measurements, published under two model identifiers, were counted as two models, which made 24 problems multi-model by duplication. With the identifiers merged it dies by its own criterion ($z = 0.88$; -1.3 points against the per-model majority; Section 4), and also by E1's original test ($z = 1.81$ against 2). Drafts v2.2 and v2.3 also called it "the only warmup signal that had survived"; H10, H16 and H18 had survived as well.

7. Section 8.1 stated that different numerical states share none of the forty generations, that the mechanism is not per-item noise, and that contention is the evaluation workers' condition. Different states share few generations but not none (30 of the 91 pairs share at least one, one pair shares 9 and two batch-1 runs share 38), and the workers' condition is not something we could observe. It also stated that the two remedies make generation bit-reproducible; they did so in three sequential runs on one card and were not tested under contention or across cards.

8. Section 8.2 described one of the three extractions as run "with the determinism flags"; only the cuBLAS workspace variable was set, not the remedy of 8.1. It stated the orientation of the official channel backwards: the regressor predicts accuracy decay, so in the orientation of its own rule the two states give 0.487 and 0.602, not 0.513 "in the expected direction" and 0.398 against it. It called the shipped threshold well calibrated on this hardware (25.6% predicted robust against a 24.4% rate), which holds in one numerical state only; in the other it predicts 33.5%. It compared these runs with an A100 figure (0.649) that came from our own extraction, was attributed to Section 5, which does not contain it, and filled a feature our extractor did not record with the exponential of the mean log-probability; with the mean top-1 probability, which the official extraction returns under greedy decoding, the figure is 0.64 (0.36 in the rule's orientation). And it did not state that E16 was not pre-registered.

9. Section 7.2 attributed 91% of DeepSeek's residual instability to the sampling seed alone; DeepSeek's A' condition, through a queueing error, also used a different prompt, so that share was not measured. The pre-registered reading grid of E13, which classifies Qwen3.5-4B as unstable by construction, had been overridden without being declared (now declared in 7.3), and gpt-oss-120b, pre-registered for labeling, was not labeled.

10. Section 7.3 described the organizers' variants as generated from symbolic reasoning chains and verified by mathematicians. The organizers' proposal (arXiv 2607.13899) does describe symbolic reasoning chains verified by mathematicians, but the published variants of the five families that 7.2 regenerates are attributed to language models in their `permutation_source` field; the sentence conflated the two and is withdrawn, and 7.3 now states that the simulated value (0.94 on average, median 0.93) is optimistic, not evidence that the label is stable.

11. Section 8.3 described the probe's permutation null as repeating the whole nested procedure; it used 60 permutations instead of the pre-registered 500 and a reduced regularization grid, and these deviations were not declared (re-run on 23 September, the reduced null reproduces its p95 of 0.574 and the full grid gives 0.570). It reported the organizers' probe artifact through an aggregation that

differs from the artifact's own inference code (0.739 against 0.709), and said that the probe reads length "from the lexical layers", which no analysis shows. It also gave the probe's gain over character count from a single fold split (+0.008 in draft v2.1, about one point in v2.2); over 20 splits the gain is about two points (0.6 to 3.2).

12. Draft v1.0 (Section 4.1) and draft v2.1 (Section 4) stated that grouped validation is what killed H1. Under an ungrouped 5-fold split, with the same code and data, H1 also dies (-0.9, -0.3 and -2.1 points over three seeds, against a bar of 2.7): grouping deepened the loss but did not cause it, and the note is withdrawn. Draft v2.1's count of dead hypotheses (nineteen) did not match its own table and is restated.

13. Section 13 stated that no evaluation data was read, inferred or probed. Aggregate facts about the evaluation sets were logged and used; they are now listed there. Drafts v1.0 and v2.1 also stated in Section 3 that we logged only aggregate counts, timings and model_id; from 31 August (v5, submission 909853) the logs also carried, per model batch, the minimum and maximum regressor score and the threshold applied, which on the two-item warmup batches are the two item scores. Drafts v2.1 to v2.3 omitted from the list in Section 13 the per-item token counts of 907967 and the fact that V8's length summaries give every item's length on batches of three items or fewer.

14. On 1 September we wrote to the organizers, as Section 7 of draft v1.0 also stated, that two of the four warmup models produced identical per-item regressor scores across four runs and that the other two visited two and three distinct states. Our logs recorded only each model's minimum and maximum score: for gemma-3-27b-it and DeepSeek, with two items each, that is every item, but for gpt-oss-120b it is two of nine and for Olmo-3-7B-Instruct two of four. The per-item statement is withdrawn for gpt-oss-120b, and the two states reported for Olmo-3-7B-Instruct are a lower bound.

15. On 30 August we wrote to the organizers, and Sections 6 and 8 of draft v1.0 repeated with a range of 30 to 40 items, that the official baseline would not finish beyond about 30 items, extrapolating from about 120 s per item on a batch of two. At the pipeline's batch size of 8 the measured cost is 26 s per item (8.1), and our five-model main-phase entry ran 21 items in 838 s.

16. Section 6.3 stated that V8 showed that length carries nothing on the evaluation set, and the abstract that it scored exactly the majority constant there. The majority constant was never measured on that set, and one evaluation on 21 and 19 items cannot show that a feature carries nothing. Section 8.3 of drafts v2.1 to v2.3 kept a cross-reference to the withdrawn reading ("Section 6.3 measured what length is worth on the evaluation set"), and PREREG_V8, PREREG_V8b and PREREG_E15 repeat it; it is withdrawn in their dated notes of 23 September.

17. Section 6 of draft v1.0 and Section 9 of draft v2.1 attributed to the organizers an explanation of the cache gap ("an outdated configuration") that they did not give, and Appendix C of draft v2.1 stated that the predictions were written before the main-phase leaderboard moved and left unchanged. The leaderboard had already moved: entries stood above the organizers' own entries on both tracks from 3 September, and on 6 September one participant's entries rose to five items above them on both tracks. P2 is anonymized.

18. Appendix B of draft v2.1, which presented itself as unchanged from v1.0, dropped the review's verdict that v1.0 had stated; neither of these two drafts said that the entry's pre-registration forbade submitting it if the review refuted it, or that the author overrode that rule. Appendix B of drafts v2.2 and v2.3 said both, but opened by stating that v1.0 had not reported the review's verdict. Appendix B now corrects both.

19. Dates. The warmup ended on 31 August 2026 (UTC; the Codabench phase record gives 15:00 UTC), not on 1 September as all drafts said. The organizers cached the four warmup models on 30 August, the day of our report, not on 31 August (all drafts: v1.0 in Section 6, which also dated our message 31 August; v2.1 in Section 9; v2.2 and v2.3 in Sections 3.2 and 9 and in Appendix B's "a container change made that same day"). The rule on training data was dropped from the website's rules on 25 August and only confirmed in Discord on 5 September (v2.1 to v2.3, Sections 7 and 9; VERIFICA_classifica_2026-09-14). The rule of the main-phase labels was published with the train set on 16 September, not 17 (v2.1 to v2.3 and PREREG_E13b). The simulation of 7.3 was first run on 13 September, not 14 (its note is misdated). And `permutation_type` was dropped from the import script on 1 September by the organizers' commit that added the Small track, not on 2 September by PR #5 (v2.1 to v2.3).

20. Related work and scope (v2.2, v2.3). The first citation named no author and presented as already documented the variation at a fixed configuration that 8.1 measures, while the cited post traces nondeterminism to kernels that are not invariant to batch size; the second used the title of the paper's first arXiv version. Section 1 also described 8.1 and 8.2 as measured "inside this benchmark's own evaluation environment", where only H18 ran inside the evaluation container, 8.1 and 8.2 ran on a VM with the workers' GPU model and image, and no measurement showed a consequence for the benchmark's labels.

21. Counting of the hypotheses. Drafts v2.2 and v2.3 said that twenty-eight entries carried a kill criterion fixed before they ran; H18's pre-registration has none, so there are twenty-seven, of which five survived (draft v1.0 had said that sixteen hypotheses, H6, H13 and H18 among them, were pre-registered each with its kill criterion fixed before execution, and draft v2.1 that all twenty-eight of its entries, T, H6, H13 and H18 included, were written down with a kill criterion). H29 to H31 were counted as pre-registered without the mark that their criterion came from the E11 plan (v2.1 to v2.3), and H2 was not declared as written after its concordance had been observed (all drafts); the table's thirty rows were called thirty-one (v2.2, v2.3); and draft v2.3 attributed the first verification to "the previous draft", which was v2.1, not v2.2.

22. Claims in the abstract and in Section 1. "Every channel we could construct" was tested (v2.1 to v2.3), but three activation channels planned in E11 were never run; the comparison with the majority constant did not name its metric (v2.1 to v2.3), and the two representation-distance statistics were scored by AUROC only; the representation distance was listed among "the statistics that pass their permutation null" (v2.2, v2.3), while it passes only the uncorrected single-statistic null and not its planned threshold; "each apparent signal" was said to trace back to something other than the model's internals (v2.2, v2.3), while our probe keeps about two points over length and the in-sample reading of the artifact was an inference; and the effect of the answer-format prompt was stated without naming Qwen3.5-4B, the only model it concerns (v2.1 to v2.3).

23. The budget guard. Draft v1.0 (Sections 1 and 8) said it makes timeouts structurally impossible, and drafts v2.1 to v2.3 called it timeout-proof, citing coverage 1.0 in every container evaluation that ran our code. No evaluation came near the limit, the guard never acted, it cannot interrupt a generation under way and relies on a fixed per-item estimate, and several of the evaluations listed did not contain it (Section 1).

24. H10 (all drafts, Section 5 and PREREG_E6_ESITO). The 0.7744 came from a logistic regression on 13 of the 14 official features, three of them constant in our extraction, not from the official regressor and threshold that PREREG_E6 specified; the substitution was not declared, the outcome file called the protocol executed without changes, and the source of the 0.692 ± 0.101 it was compared with (the organizers' proposal, a different protocol) was not cited.

25. H13 (v1.0 §5.2, PREREG_E8 and the notes added to PREREG_E7 and E8, and all later drafts in Sections 1, 4 and 5). The oracle's +7.3 was attributed to its freedom to degenerate into a constant. Excluding constant predictions it still gains +5.5; all of its gain comes from thresholds chosen on the items they are scored on, and the fall to -2.8 comes from fixing the number of predicted positives, to which Llama contributes nothing.

26. H14 (all drafts) was described as a realizable calibration via public priors. Each target's public rate is its rate on the tested items, so H14 coincides with the oracle-k ceiling of E8 (both 0.7431); its verdict stands.

27. Warmup rows changed in draft v2.2 without being declared: H13 moved from dead to a withdrawn ceiling, H16 and H17 from survives to within noise and not established (one of three re-executions of H17's bundle scored 9/17, which meets its kill criterion), H17 had been described as a per-model threshold calibration (only the gpt-oss threshold was calibrated) and H3 as 560 ordered pairs (56 ordered model pairs, 560 predictions), and H6 and T were marked as having no kill criterion.

28. "All validation is grouped" (all drafts). The cross-model tests (H7, H12 to H15) hold out the model but not the problem: 234 of 335 test rows in H7 and 70 of 109 test pairs in H12 to H15 have their problem in training. No verdict changes (Section 4).

29. H1, H7 and T (all drafts) and H8 (v2.2, v2.3) were computed on the 335 published pairs, counting the 52 duplicated DeepSeek measurements twice. Counted once, H1 is -7.4 (not -8.7), H8

+1.8 against the global constant (not +1.5), T +7.1 pooled (not +6.0), and H7 stays at +0.0; no verdict changes.

30. E11 (v2.2, v2.3). The statement that "the E11 analysis applied Holm to two tests instead of five" is false: the analysis applied no multiplicity correction, and H30 would have failed Holm over its two tests (permutation p 0.030 against 0.025 at alpha 0.05). Further departures from the plan were not declared: the confirmation set was not held out, the AUROCs were computed on the raw scalars with the orientation chosen on the data, the text baseline of H28 was reduced to character count, the stratified and bookkeeping reference AUROCs that the plan made obligatory were not reported, and only the primary depth was extracted (Section 4).

31. The V4 tie was presented as caused by re-execution drift in v1.0 (Section 7 and Appendix B) and v2.1 (Appendix B: "thereby exposed the nondeterminism that Section 8 later explained"), and as "the first sign of the re-execution noise" in v2.2 and v2.3 (Section 4), although none of the three explanations recorded at the time was verified; v1.0, v2.2 and v2.3 also quoted its pre-registration without its exception (a tie was impossible unless the DeepSeek path failed). Draft v1.0 (Section 7) and drafts v2.2 and v2.3 (Appendix B) also said that the tie led us to measure re-execution drift; the pre-registration of that measurement (PREREG_V7) gives as its reason the drift of regressor scores between the logs of v5 and v6 (Appendix B).

32. The labeling rule. Drafts v1.0 and v2.1 stated that it is exactly recoverable from the labeled rows, and v2.1 that the majority rule of the organizers' proposal agrees with them on only 221 of 379 rows. The labeled rows do not identify the rule (a majority of correct perturbed answers also agrees 379/379), and the 221 is reproduced only by comparing a count of variants with a count of predictions. Draft v2.1 also stated that no public labeled item exists for any of the other four main-phase models, while gpt-oss-120b has ten public pairs, five of them with a declared label, and it called four labeled files aggregated where three are. No draft cited FAQ 7, and none gave the source of the proposal's majority rule (arXiv 2607.13899, its Section 1.4): v2.1 mentioned that rule only through the wrong 221 of 379, and v2.2 and v2.3 dropped the mention instead of correcting it (2.1).

33. The ground truth (all drafts). The 164 DeepSeek labels were presented as the organizers' labels; 26 of them carry no declared label and were derived by us with the conjunction, and a majority rule, equally consistent with the declared labels, would call 21 of those 26 robust (2.2). Draft v2.3 also stated in 2.2 that the main-phase train set is not used in its analyses, while its 6.4 evaluated our entry on it.

34. The organizers' sampling regime (v2.1 to v2.3, PREREG_E13, PREREG_V8). "About ten variants per family, one sampled prediction per variant, ten sampled predictions on the base problem" describes the AIMO-problem file, not the files behind the DeepSeek labels, where most variants were answered ten times and the base between 1 and 60 times; Section 7.2 said the regeneration used "the organizers' sampling counts" (v2.2, v2.3) and was titled "regenerated at their scale" (v2.1 to v2.3) (2.1, 7.2, 7.3).

35. Section 2.3(b) of drafts v2.2 and v2.3 gave the official feature's range as 0.959 to 0.999 and the share inside the split range as 91%, with the thresholds rounded to [0.968, 0.993] as in v2.1; the minimum is 0.958, and between the true lowest and highest thresholds (0.9684 and 0.9935) the share is 92.6% (98.8% on DeepSeek).

36. `Force_Latest_Multiple` (v2.1 to v2.3, OSSERVATO_MAIN of 2 September, PREREG_V8, PREREG_V8b). We read the declared leaderboard rule as keeping only each participant's latest submission per task and justified the resubmission 925822 by it. Under this rule Codabench adds every scored submission as a row of its own and removes none; the valid row 924969 had stayed on the leaderboard, and the resubmission is unaffected (6.3).

37. V8's parameters. PREREG_V8 and the bundle's docstring call 98 the median of the character counts of the 174 public problems, and draft v2.1 calls it the median character count; it is the lower median (the median is 98.5). The pre-registration and the docstring give the rule's balanced accuracy on public data as 0.658, the figure for a threshold of 99; drafts v2.1 to v2.3 and a note of 23 September in PREREG_V8 gave 0.653, itself a double rounding: for the shipped threshold of 98 it is 0.652 over the 335 published pairs and 0.631 over the 283 distinct ones. Drafts v2.2 and v2.3 also said that the rule uses no training data; its threshold and direction were fixed on the public datasets (6.3, Section 13).

38. Other statements in Section 6. Drafts v2.2 and v2.3 said that whether 11/21 and 10/19 is the majority class "was not measured", omitting that on 21 and 19 items the majority constant scores at least 11/21 and 10/19, so neither V8, nor v6, nor the organizers' entries scored above it. Drafts v2.1 to v2.3 described "a five-model, ten-minute pipeline", while the runs took 14 minutes with five models and 8.5 minutes with four; and draft v2.3 did not update the scope sentence of 6.1 when it added 6.4.

39. Section 7.2. The 638 example variants (v2.1 to v2.3) are 624 in the five families (the rest are expert-written or of a sixth family), of which our pools used 621. The statement that on Skywork and Olmo most of the instability comes from which variants are drawn (v2.2, v2.3 and the abstract) holds under the earlier rule for Skywork only: on Olmo the seed alone accounts for 56% of it. The subsampling curve's concordance "does not improve" (v2.1 to v2.3), while it rises from -0.02 to $+0.03$. Draft v2.1 and PREREG_E13 stated that raising the base-accuracy threshold to 10/10 changes nothing, and drafts v2.2 and v2.3 stated the property behind the unchanged kappas without limiting it to draws A and B. And the table's rate column, draw A over 164 problems, was not labeled as such (v2.1 to v2.3).

40. The solver and the answer key (v2.1 to v2.3). The "independent solver" of 7.2 is gpt-oss-120b itself at a higher reasoning effort, independent of the labeled models but not of the generator; its answer on the original is also the answer key against which every labeled answer was scored, which no draft stated; and the translator of H22 (microsoft/phi-4) was not named.

41. The simulation of 7.3. Draft v2.1 said that it reproduces the organizers' robust rate, calling 13.8% of the 116 pairs declared (as does VERIFICA_classifica_2026-09-14); it is the share robust under the stricter reading the simulation uses. Draft v2.3 described its data as three per-family files, while one of the three, `val-sample`, is aggregated and restates the first (721 rows are distinct); it said that its 16 robust pairs were robust "under the rule", while 30 are robust under the rule of 2.1 and 16 under that stricter reading; and its abstract and Sections 11 and 12 gave 0.93 where its 7.3 gave 0.94 on average. Drafts v2.1 and v2.2 called the value an upper bound and draft v2.3 called it optimistic, all without the control, run on our own labels, that shows by how much it overstates (about 0.4). No draft said that the simulation covers six model identifiers besides DeepSeek and the organizers' expert families.

42. Section 7.4 and R3. The headline 80% (v2.1 to v2.3) came from the 30 problems of the feasibility check; on the full runs the share is 0 and 145 of 164 (88%). "DeepSeek-R1 closes spontaneously either way (9.0 of 10)" (v2.1 to v2.3) gave the bare-text figure for both prompts (9.0 and 9.2). Drafts v2.1 to v2.3 also said that Qwen3.5-4B "never closes in \boxed{}" with the bare text, while it did in 5 of the 1,640 base samples of draw A and 8 of the 1,640 of draw B. And R3 said that the prompt, as far as we could find, is not published (v2.3; v2.1 and v2.2 only asked for it to be published), without mentioning the solve prompt in the organizers' baselines repository.

43. Section 7.5 (v2.2, v2.3). "Two of our three predictions were wrong": none held in full, since the third predicted a concordance between 0.10 and 0.40 and it came out at 0.02. The old and new kappas were compared on different sets of problems without saying so, and without their n; on the same problems the change of rule adds 0.01 to 0.11. The gap to the official labels was called one of severity, while our robust rate is the same in both of their classes and our max_drop does not correlate with theirs. The statement that with the same variants and a new seed agreement is high on Skywork and Olmo, so that the label depends mainly on which variants are drawn, holds for Skywork only: on the 126 problems of Olmo labeled in all three series the seed accounts for about half of the instability. The concordance's upper bound is 0.13, not 0.14; the DeepSeek-identifier rows are 54 on 52 problems; the prediction was 0.60 on three of four models, not on every model; and the abstract's 71 to 75% did not name the rule it holds under.

44. Section 8.1 (v2.1 to v2.3). The failed predictions P-E14d and P-E14e were not reported, and the pre-registered measure m4 was not computed; "each configuration has its own fixed point" omitted that the two deterministic configurations ran on different cards; crossing counts from different numbers of runs were compared (and R1 set 32 over six runs against 24 over five); and Section 12 named compilation as the inferred cause, which the runs do not single out.

45. Section 8.2. Its title ("not stable in sign", v2.1 to v2.3), its statement that one state orders against the rule and the other with it, and the abstract's "this drift decides whether the official baseline's score channel carries any signal" (v2.2, v2.3) claimed more than the data show: one state is at chance, the other weakly above it, and their difference is not significant.

46. Section 8.3. The artifact's stored scores were called out-of-sample scores (v2.1 to v2.3); they are validation balanced accuracies of a rule that almost always predicts one class. Probable in-sample training was offered as the explanation of the artifact's AUROC (v2.1 to v2.3), while its margin tracks problem length and its control probes, trained on shuffled labels, give 0.68 by themselves. "The probe reads mostly, though not only, problem length" (v2.2, v2.3) claimed more than the control shows, which is redundancy as predictors. The per-layer maximum is 0.5475, not 0.547 (v2.1 to v2.3); Section 4 said the failed prediction of E15 is reported in 8.3, which did not report it (v2.2, v2.3); and the single-split triple was printed in an order that contradicted its sentence (v2.3).

47. Section 8.4 (v2.1 to v2.3) did not state that its rates apply the DeepSeek regressor to every model, as our entry does and the official inference does not, nor that gpt-oss-120b is missing because its extraction failed; the introduction of Section 8 called this the unmodified official path.

48. Compute (v2.1 to v2.3). The Modal total was given as about 27 GPU-hours, a figure no file supports; the dated files document about 12 to 13. The VM total was given as about 100 card-hours, the estimate for E13 alone.

49. Correspondence. The status of the results was dated 21 September in a draft of 23 September (v2.3), and the note on 8.1 that the E14 pre-registration planned to send by 22 September was not mentioned (v2.2, v2.3); the VM was said to have run Section 8 only (v2.1 to v2.3), while E13 ran there too; E16 was presented as the only experiment on the VM without a kill criterion (v2.2, v2.3), while E13 and E14 had none either; the VM was said to be released "at the organizers' request" (v2.2, v2.3), while they asked whether we still needed it; the organizers were said to have raised the cost of averaging three runs in the final evaluation, and our reply to have said that three runs stabilize the published number (v2.1 to v2.3), while their worry concerned running each Codabench evaluation three times during the phase and our reply said "substantially steadier" (Section 9); and R4 said the organizers were "considering" more rows for Skywork (v2.3), while they wrote that the train set can be expanded.

50. Record-keeping (v2.2, v2.3). Section 12 of v2.2 and Section 1 of v2.3 said that three analyses had been run from scripts that were not saved; the cross-run analysis of E14, the figures of E16 and of 2.3b, and the balanced accuracy of 6.3 were in the same position. All are now computed by saved scripts (Section 12).

51. Section 13. The pre-registrations were described as all carrying kill criteria (v2.1 to v2.3; E13, E14 and V7 carry none), and v1.0 (in its Section 11) listed PREREG_E1 through E8 and v2.1 PREREG_E1 through E16, although E5, E11 and E16 have no pre-registration file; the deviations declared for the first time omitted E4, E6, most of E11, two of E13 and E14 (v2.2, v2.3), and the list of deviations preserved in place omitted V4 (v2.1 to v2.3); "no other logged aggregate was used to tune a submission" omitted H17's motivation and the budget guard's estimate (v2.2, v2.3); the disclosure of AI assistance omitted the bundle code, the VM, the uploads and most of the records it read (v2.2, v2.3); and the release of every dated file did not provide for redacting other participants (v2.1 to v2.3).

52. Appendix C and Section 6.3. P2 was restated about a single entry (v2.1: "the entry that led the leaderboard on 7 September"; v2.2, v2.3: "the entry that led both tracks"), while the dated prediction concerns a participant whose leading entries on the two tracks are distinct submissions. P-V8 was restated without its conditions and P-V8c was omitted (v2.1 to v2.3; v2.1 also omitted the alternative of about 0.66), and Section 6.3 gave only the band of P-V8a, without its condition, as the prediction (v2.1 to v2.3). Two of the three predictions of PREREG_V8b were omitted (v2.2, v2.3), and the organizers' entries were called baseline entries without a source (v2.2, v2.3).

53. Section 11's closing sentence (v2.1 to v2.3; v2.1 excepted nothing, v2.2 and v2.3 only P2) did not except P1 and the public leaderboard figures cited in the report, which concern participants' scores as well. Section 9 of drafts v2.2 and v2.3 also contained a statement about other participants' activity in repository issue #6, which the closing sentence did not except either; it is withdrawn, and Section 9 now says only that the earlier wording was still quoted there on 16 September.

54. Statements corrected in place without an item until this draft. Draft v1.0 said that the A100 extraction reproduced the organizers' pipeline, with torch and transformers pinned to the container's versions; it was our own re-implementation of the extraction, which differs from the official code on one feature and omits another (2.3b), and torch was 2.13.0, the container's 2.12.1 (Section 5). Draft v2.1 said that every number had been recomputed from disk before being written (several were not,

Section 12); that a representation-distance signal at AUROC 0.577 had been exposed as prompt length by the neutral-text control (0.577 is the raw secondary statistic, below its own null; the tested statistic, H30, gave 0.615 and lost to the character count, Section 4); that the gap between the two gpt-oss thresholds was a first sign of the drift of Section 8 (6.2); that the container's main-phase score ranges are higher than the public ones on every model, so that a threshold calibrated on public problems does not transfer (they fall inside the public distributions and cannot tell, 8.4); that Section 8.2 found the 0.7744 of Section 5 not to survive (it was not repeated there, Section 5); that batch size 1 costs 182% and 73.0 s per item (178% and 72.3 s on average over its three runs, 8.1); that the second physical card reproduced the first card's state exactly (two of its three runs did, and the third landed in a state of its own, 8.1); that the invalidated Qwen runs took thirteen GPU-hours (about fifteen card-hours, 7.4); and that no other process could regenerate the declared families to useful agreement (7.3, R4). Drafts v2.1 to v2.3 called the regenerated rule the label's stated definition (abstract, Section 7), called our perturbations and sampling variance orthogonal to what the label measures (7.1), described the aggregation as keeping only the detrimental type of a non-robust item, while it keeps the type with the lowest perturbed accuracy and drops milder detrimental ones (2.3a), and described the gpt-oss calibration set as ten public items (v2.1: "10 public gpt-oss items") without saying that three of the rows are of a fine-tune and carry its only robust label (6.2). All drafts called `val-sample` test-matched (abstract, Sections 1 and 4), and drafts v2.2 and v2.3 attributed to the organizers a description of it as covering all test model types, which we have found only in our own dated note (2.4). Drafts v2.2 and v2.3 called the new thresholds the rule the organizers now use and said that the regeneration ran under both of the organizers' labeling rules, while the conjunction is our reading of the data (abstract); said that grouped validation had never been compared with an ungrouped split (Section 4 and item 12), that the organizers' variants break the models more often than ours (only DeepSeek can be compared, 7.3), and that the organizers' labeling prompt and base-accuracy threshold had not been published at the time (they still have not, 7.2; a solve prompt is in their baselines repository, R3); and listed only 8.2 and 8.3 among the measurements that were not pre-registered. Drafts v2.1 to v2.3 also concluded that a learnable target can come only from the organizers (v2.1: "the only route to a learnable target runs through the organizers' own variants"; v2.2: "a learnable target has to come from the organizers' own variants"; v2.3: "the labeled split is the only learnable target"), which our failed regeneration does not show (R4).

D Limitations, deviations, data provenance and compute

Carried over verbatim from the full report (draft v2.4, 23 September 2026), which is released with this report. Section and appendix numbers in this appendix refer to that report; Appendix E maps them to this one.

Its paragraph on the use of AI assistants is superseded by the AI usage statement of this report, which records a second exception (submission 1016734).

Full report, Section 12: Limitations

Ground truth under the earlier labeling rule exists for two evaluated models: 164 pairs for DeepSeek, 26 of whose labels we derived (2.2), and 10 for gpt-oss-120b at low reasoning effort, five of them with a declared label. Every probe and every concordance in Sections 7 and 8 is measured on DeepSeek and assumed, not shown, for the other four. The main-phase train split (2.1) labels all five evaluated models under the new rule, but with only 3 to 37 labeled rows per model, and apart from the evaluation of our entry and the five-row comparison in 6.4 the analyses in this draft do not use it. Sections 8.1 and 8.2 use 40 and 174 problems and one model. The fourteen-state structure is established; its cause is inferred, not proven: a choice of kernels or algorithms made when the process starts, which either disabling compilation of the static-cache forward or forcing deterministic algorithms removed in our sequential runs. The remedies were tested only in sequential runs, each on one card. The simulated 0.94 in 7.3 treats observed frequencies as true probabilities, and on our own labels the same procedure overstates the measured reliability by about 0.4. The regeneration in 7.2 used gpt-oss-120b prompted with the public examples and gated by itself; the public records attribute the organizers' own variants of the same families to language models as well, so the negative concordance is a fact about our generation, gating, answer key, sampling counts, decoding and prompting, not about the families.

Several analyses were first run from scripts that were not saved in the project. On 23 September three were re-created and saved: the subsampling curve of 7.2 and the second-level length control of 8.3, both extended from one draw or split to twenty (the second gives a gain of about two points, against the 0.8 of the single split first reported), and the simulation of 7.3, re-created from its dated description. The same day, scripts were written and saved, with their outputs, for the cross-run analysis of E14 (8.1), the AUROCs, rates and accuracies of E16 (2.3b, 8.2, 8.4), the controls of the public data (Section 2, 6.3), the artifact controls and the nested null of 8.3, the recomputations of the warmup hypotheses (Section 4), the permutation p-values of E11 and the other controls the second and third verifications required (Section 13). A few figures still rest only on a dated record: the ICC values of H29, written to a Modal volume that was not downloaded; facts read from container logs at the time and recorded in dated files (the item counts per model of 3.2 and 6.1, the 119 s per item of 907677, V8's per-item predictions on the current tasks, v6's runtimes and gpt-oss thresholds, and per-model score ranges); Codabench scores and leaderboard facts recorded in dated files and absent from our saved snapshots (the warmup scores of 907310, 907677, 909843, 909853, 909994, 910002 and 910013, V8's scores of 13 and 14 September on the original tasks, and the leaderboard figures of 29 August and of 13 and 14 September); and the compute totals of Section 13. Nothing here estimates performance on the final held-out problems; Appendix C states what we predicted before knowing.

Full report, Section 13: Reproducibility and disclosure

Pre-registrations. PREREG_E1 to E4, E6 to E10, E12 to E15 and E13b, and V4 to V8b, each dated before execution. The hypothesis tests among them carry kill criteria; E13 (a reliability measurement read through a pre-registered grid), E14 and V7 (measurements with written predictions) carry none, and V4, V8 and V8b state predictions and decision rules. Declared deviations (E6, E12, E13, E14, E15, and V4, whose entry was submitted once as a measurement although its pre-registration said a refuted entry would not be sent, Appendix B) are preserved in place, and dated correction notes were appended on 21 and 23 September. E5 is documented only in its outcome file; E11 ran from a written plan (PIANO_E11_definitivo) without the frozen pre-registration it called for; E16 (8.2, 8.4) was not pre-registered, although our reply of 7 September to the organizers said every experiment on their VM would be. Deviations declared here for the first time: E4 (H7 compared with the global prior of the training models instead of the pre-registered per-model majority, against which it scores -1.2 points); E6 (H10 run as a logistic regression on 13 of the 14 features in place of the pre-registered regressor and threshold); E11 (three planned primary tests not run; a Holm correction planned over five tests at alpha 0.010, announced over two and not applied; the confirmation set not held out; AUROCs computed on the raw scalars with the orientation chosen on the data; the text baseline reduced to character count; the stratified and bookkeeping reference AUROCs not reported; only the primary depth extracted); E13 (gpt-oss-120b not labeled; DeepSeek's A' condition labeled with a different prompt; the per-family subsampling curve replaced by a per-problem grid, 7.2; the pre-registered reading grid not followed, 7.3); E14 (measure m4 not computed); and E15 (in the nested null, 60 permutations instead of 500 and two of the four regularization values, the latter re-run with the full grid on 23 September; Holm over the two primaries not computed).

What we logged, and what we know about the evaluation sets. Container measurements: submissions 907310, 907677, 907967, 909567, 909843, 909853, 909976, 909994, 910002, 910013, 912096, 912097, 924968, 924972, 925822, 936802, 936806. They logged aggregate counts, the number of items per model, per model batch the minimum and maximum regressor score and the threshold applied (v5 to v7), per-model length summaries (V8: minimum, median and maximum character count, which on batches of three items or fewer are every item's length, unattributed), timings, model identifiers and, in 907967, the GPU, memory and library versions and the prompt and generated token counts of one DeepSeek warmup item; we read those logs. No problem text and no per-item label from the evaluation sets was logged, stored or recovered. Aggregate facts about those sets were observed or inferred and are reported where used: item counts per model (3.2, 6.1), the class balance implied by the pre-registered null probe and by constant entries (3.1, 6.3) and per-model score ranges (8.4). One warmup submission was built on aggregate facts about the evaluation set and is disclosed in Appendix B. Logged aggregates also shaped two other design choices, neither of which used evaluation labels: H17 (v6) calibrated a separate threshold for gpt-oss-120b because its logged scores on the warmup set (9 of 17 items, 0.340 to 0.870) straddled

the inherited threshold, while the threshold itself was fitted on the ten gpt-oss rows of `val-sample`, one of them robust (6.2); and the budget guard's per-item cost estimate was set from container timings (130 s after the measured 119 s, then 90 s after the 21 to 59 s logged by v5).

Compute. GPU time on Modal: the dated files document about 12 to 13 GPU-hours and about 34 dollars, of which 9.2 A100-hours (about 19 dollars, funded personally) went to the extraction of Section 5 and the rest to its pilot and to E9 to E12; the billed total, including start-up and loading time and how much of it platform credit covered, will be itemized from the billing records before release. On the organizers' VM, E13 took about 100 card-hours by the estimate in its outcome file, and E14 and E16 several more; a total will be itemized from the run logs before release.

Release. Code, extraction outputs (hashes, scores, per-variant answers), the analysis scripts and every dated file will be released with the final version, including drafts v1.0 and v2.1 to v2.4 (draft v2.0, of 14 September, was edited in place into v2.1 and was not kept as a separate file; its four corrections are items 1 to 4 of Appendix D). In the dated files, the names, handles and personal details of other participants, and every passage about their scores, methods or conduct, will be redacted, with each redaction marked, as in Appendix C. The invalidated Qwen labeling runs are kept with a file stating why. The analysis scripts saved on 21 and 23 September are `controllo_h2_alias_2026-09-21.py`, in the project root (H2 with the duplicated rows counted once, Section 4), and, in `vm/`: `e13b_regola_nuova.py`, `e13b_controllo_modello.py`, `e15_percorso_ufficiale.py`, `simulazione_affidabilita_organizzatori.py`, `e13_sottocampionamento.py`, `e15_controllo_lunghezza.py`, `e15_null_nidificato.py`, `e14_tabella.py`, `e16_analisi.py`, `e15_artefatto_controlli.py`, `e11_holm.py`, `warmup_ricalcoli.py`, `dati_pubblici_controlli.py`, `controlli_verifica2.py`, `controlli_verifica3.py`, `controlli_verifica4.py` and `v8_train_main_v2.py`, with their outputs in `vm_risultati/`; the Codabench snapshots of 7, 21 and 23 September and the three files downloaded from the organizers' baselines repository (`prompts/solve.txt`, `data/math-robust-agg.csv` and `scripts/probe.py`) are in `dati/`.

Reproducing our score. Our entry on the current tasks is V8. The Main bundle `solution-v8-main.zip` (SHA-256 `7b70f15c51752dcf4285e477d58c541d0ecf052d833c325c0702dbbd054ffb5f`) contains a single `solution.py` that defines `are_robust(model_id, problems)`, returns `[len(p) >= 98 for p in problems]` and logs only per-model counts and length summaries; the Small bundle `solution-v8-small.zip` (SHA-256 `0b83b55f94dc16b9852a7c2b35d71ac5787e43d81afd6396dee00fb5e011e500`) adds the empty marker file `small.txt`. Upload each bundle with only its own task selected: Main on task 36237 (expected 10/21, submission 936802), Small on task 36241 (expected 7/14, submission 936806). The rule loads no model and reads no data at run time. Its two parameters were fixed on the six public `aimo-interp` datasets, which are its training data: the threshold, 98, is the lower median of the character counts of their 174 distinct problems (the median is 98.5), and the direction, longer problems robust, is the one with the higher balanced accuracy on their labeled pairs (0.652 over the 335 published pairs, against 0.348 for the reverse). `vm/controlli_verifica3.py` recomputes the threshold and `vm/dati_pubblici_controlli.py` the balanced accuracies.

Use of AI assistants. This work was carried out with an AI assistant (Claude, by Anthropic, used through Claude Code). It drafted the pre-registrations, including their hypotheses, predictions and kill criteria; wrote the solution code of our submission bundles and the analysis and extraction scripts, and ran them; operated the organizers' VM over SSH for E13 to E16; made most of our Codabench uploads through the author's browser, at the author's instruction (recorded in PREREG_V4 for 909843 and in PREREG_V8b for 936802 and 936806), except the resubmission 925822 of 14 September, which it made without a specific instruction, to restore what it had wrongly taken to be a lost Main row (6.3); read the public Codabench, GitHub, Hugging Face and competition-website records, the organizers' Discord channels, their proposal on arXiv, the works cited in Section 1 (on arXiv, the NeurIPS website and the Thinking Machines Lab website) and the author's email correspondence with them, behind the dates, quotations and leaderboard figures of the header, of Sections 1, 2, 3, 6, 7, 8, 9, 10 and 13 and of Appendices B to D; drafted the correspondence and this report; and ran automated multi-agent adversarial reviews of hypotheses, drafts and code, including

the verifications behind items 5 to 54 of Appendix D. The author directed the experiments and the submissions, with the exception stated, and is responsible for every claim.

E Section map between the full report and this report

The full report (draft v2.4, 23 September 2026) is released with this report. Its sections correspond to this report as follows; the sections marked as not reproduced are only in the full report. Sections 2 to 7 of this report are condensed from the nine-page text given in Appendix J under the same numbers.

Full report (draft v2.4)	**This report**
Abstract; 1. Stance and contributions	Abstract; Section 1
2. The instrument from the outside: public data	Section 2 (condensed; 2.3(b), the withdrawn regressor defect, is not reproduced)
3. The instrument from the inside: warmup measurements	Not reproduced; H16 to H18 in Appendix A
4. The falsification campaign	Section 3 (condensed) and Appendix A (verbatim)
5. GPU-scale replication (warmup)	Section 3 and Appendix A (H10, H13)
6. The main phase: the entry that measures, and the entry that does nothing	Section 6; predictions in Appendix B
7. Can the label be regenerated from its definition? (introduction)	Section 5, first paragraph; the rule on training data in Appendix H
7.1 Our families, translation, self-consistency: three dead ends with ground truth	Section 3 (Table 1)
7.2 The organizers' own families, regenerated; 7.3 What this does and does not show	Section 5 (Table 2), 5.1 and 5.2; the first run of 7.2 (E12, H32) in Section 3 (Table 1)
7.4 The answer-format dependence	Section 5.3
7.5 The same data under the main-phase rule (E13b)	Section 5.1 and Table J.3
8. The instrument on its own hardware (introduction)	Section 4, first paragraph; compute in Appendix D
8.1 The structure of re-execution noise	Section 4 (Figure 1) and Table J.2
8.2 The official score channel does not reproduce its ordering between identical runs	Section 4, last paragraph (the comparison with our A100 features is not reproduced)
8.3 Linear probes on the residual stream, ours and theirs	Section 3, Length and probes
8.4 Per-model score distributions on this hardware	Not reproduced
9. Correspondence with the organizers, and the compute grant	Not reproduced, except the thanks (Acknowledgments), the question about measurements on the organizers' machine (Limitations and Appendix K) and the rule on training data (Appendix H)
10. Recommendations	Section 7, renumbered: R1 to R3 keep their numbers, R5 is R7 and R6 is R5; R4 (enlarge the labeled training split) is not reproduced; R4 and R6 of this report are new
11. Corrections since draft v1.0	Appendix C
12. Limitations	Limitations (after Section 7) and Appendix D
13. Reproducibility and disclosure	Appendix D and Appendix K; the use of AI assistants also in the AI usage statement
Appendix A. Floor algebra	Appendix F
Appendix B. The allocation episode	Appendix G
Appendix C. Pre-registered predictions to be scored against the final evaluation	Appendix B
Appendix D. Corrections to our earlier statements	Appendix C

F Floor algebra

Carried over verbatim from the full report (draft v2.4, 23 September 2026), which is released with this report. Section and appendix numbers in this appendix refer to that report; Appendix E maps them to this one.

With floor f, raw score s maps to capability $\kappa = (s - f)/(1 - f)$ and raw error is amplified by $1/(1 - f)$. Knowing κ to within ε needs $n \geq 0.25/(\varepsilon(1 - f))^2$ items.

G The allocation episode

Carried over verbatim from the full report (draft v2.4, 23 September 2026), which is released with this report. Section and appendix numbers in this appendix refer to that report; Appendix E maps them to this one.

Corrected from v1.0, which reported the review's conclusion but not that our pre-registration made a refutation a reason not to submit, nor that the author overrode that rule, and which presented the exclusion from the leaderboard as the review's consequence; draft v2.1 did not state the verdict at all (Appendix D, item 18). One aggregate-informed entry (909843) was pre-registered with the rule that it would not be submitted if our adversarial compliance review refuted it. The review refuted it: all of its expected advantage came from observed statistics of the evaluation set. The author overrode the rule and submitted it once, as a measurement, kept off the leaderboard. It tied the entry it modified, which its own algebra ruled out unless the model path on the DeepSeek items failed. Of the three explanations recorded at the time (numerical drift on the DeepSeek items, a change in the evaluation set with the container update the organizers reported on 31 August, a changed label total), none was verified then; the cache update had preceded both entries, since the getting-started README listed all four models as cached from 30 August. The measurement of re-execution drift (H18; Section 8) was prompted by the drift of regressor scores between the logs of v5 and v6, at equal code and inputs, noticed on 31 August, after the tie (PREREG_V6), which is the reason the pre-registration of the repeated runs gives (PREREG_V7); drafts v1.0, v2.2 and v2.3 attributed it to the tie (Appendix D, item 31).

H Assets and licenses

New in this report; licenses as stated on the Hugging Face and GitHub pages on 23 October 2026.

Models. gpt-oss-120b (OpenAI, Apache-2.0); Olmo-3-7B-Think (Allen Institute for AI, Apache-2.0); DeepSeek-R1-0528-Qwen3-8B (DeepSeek, MIT); Qwen3.5-4B (Qwen, Apache-2.0); Skywork-OR1-Math-7B (Skywork; no license is stated on its model card); phi-4 (Microsoft, MIT), used only for the translations of H22; and gemma-3-27b-it (Google, Gemma Terms of Use) and Olmo-3-7B-Instruct (Allen Institute for AI, Apache-2.0), which our warmup entries loaded in the organizers' container for H16 to H18. The model revisions on the organizers' virtual machine were Qwen3.5-4B 851bf6e, Skywork-OR1-Math-7B fcef970, Olmo-3-7B-Think d97e442, DeepSeek-R1-0528-Qwen3-8B 6e8885a and gpt-oss-120b b5c939d (as recorded on 17 September).

Data. The organizers' datasets published under `aimo-interp` on Hugging Face: `val-sample`, `augmented-sample-math`, `augmented-sample-math-full-filtered`, `augmented-sample-math-agg-filtered`, `aimo-interp-challenge-sample-full` and `augmented-sample-math-agg`, whose cards state no license, and the main-phase train set `train-main-v2`, whose card states the license as "other" without further detail; we used its revision 23791d9 of 17 September 2026 (82 rows, 71 labeled), which the organizers extended to 94 rows on 15 October. The problems of the six earlier datasets come from the MATH dataset [7] (MIT license) and from the AIMO 2025 reference problems, as their `dataset_id` field records (`hendrycks-math-train-reference-500`, `hendrycks-math-test-reference-100`, `aimo-2025-reference`). The website's rule against extra labeled data was commented out on 25 August (the Codabench Terms kept it until at least 16 September) and retired explicitly on 19 September [4]; on 5 September the organizers wrote in Discord #qa that participants can use the perturbations and labels they generate themselves "for anything" [6]. On 2 October, answering a question in repository issue #10, they allowed the publication of self-generated prompts, labels and

output-level metrics, and of per-example rows derived from `augmented-sample-math-agg` without the problem text or the models' raw outputs [5]. Our perturbed problem texts go beyond that exchange; on 2 November, answering the author's email, the organizers allowed their release.

Code. The organizers' repositories on GitHub. `aimo-interp/getting-started` (no license stated) contains the evaluation harness, the official uncertainty-feature extraction (`uncertainty_profile`) with the shipped DeepSeek regressor, and the DeepSeek probe artifact with its inference code; we used the last three [5]. Our model-running bundles include the extraction code and the regressor unchanged; on 2 November the organizers allowed their redistribution, and we release the bundles as submitted. `aimo-interp/baselines` (Apache-2.0) contains the organizers' encoding and probe training code; we used three of its files, `scripts/probe.py`, `prompts/solve.txt` and `data/math-robust-agg.csv`. vLLM [8] (Apache-2.0) ran the generation, gating and labeling of Section 5 (version 0.28.0 on the organizers' virtual machine, as recorded on 17 September).

I Prompts

New in this report; taken verbatim from our scripts `vm/e13_varianti.py` and `vm/e13_etichetta.py`, with the placeholders in angle brackets.

Generator. gpt-oss-120b at low reasoning effort, temperature 1.0, top_p 1.0, at most 4,096 tokens, one request per problem and family. For each draw, problem and family, five examples of that family are drawn reproducibly from the public examples (Section 5), excluding the problem itself. Lines in parentheses describe the prompt rather than quote it.

```
You create variants of mathematics competition problems for a robustness
benchmark.
Perturbation type:  <family>.  <family instruction>
The variant must be a complete, self-contained problem with exactly the same
final answer as the original.  Do not solve the problem.  Do not add hints
or solutions.

Here are 5 examples of this perturbation type:

Example 1
Original:  <public original>
Variant:  <public variant>

(Examples 2 to 5 in the same form)

Now write 10 different variants of the following problem.  Make them
genuinely different from each other.  Output each variant between <variant>
and </variant> tags, and nothing else.

Original:  <problem>
```

The family instructions:

- `rephrase`: Rewrite the problem in different words while preserving exactly its mathematical content and its final answer.
- `rename`: Rename the variables, people, objects or entities in the problem while preserving its mathematical structure and its final answer.
- `domain`: Transfer the problem to a different real-world surface domain (a different story, setting and objects) while preserving exactly the same mathematical structure and the same final answer.
- `distract`: Insert irrelevant but plausible additional information (extra facts, numbers or context) that does not change the problem or its final answer.
- `typos`: Introduce harmless typographical or formatting noise (misspellings, spacing, punctuation, capitalisation) without changing the meaning of the problem or its final answer.

Gate and reference answers. gpt-oss-120b at medium reasoning effort, temperature 1.0, top_p 1.0, at most 8,192 tokens, with this instruction line followed by a blank line and the problem or the variant:

```
Solve the following problem.  Reason step by step, then give only the final
answer inside \boxed{}.
```

Labeling. The same instruction line, followed by a blank line and the problem or the variant, was given to Qwen3.5-4B, Skywork-OR1-Math-7B and Olmo-3-7B-Think in all their series and to DeepSeek in A′; DeepSeek received the bare text in A and B. Sampling: temperature 0.6, top_p 0.95, at most 16,384 tokens; ten samples on the base problem and one on each variant.

J Extended version of Sections 2 to 7

This appendix gives Sections 2 to 7 in the longer form of our nine-page draft of 22 October 2026, which the main text condenses; section numbers are those of the main text. Tables J.1 to J.3 give in detail what the main text condenses into Table 1, Figure 1 and Table 2 (Appendix A gives every entry of Table J.1). Two changes from the nine-page draft: in Section 6 the resubmission of 28 October and our nominations for the final test replace two placeholders, and in Sections 5.1 and 5.2 the comparisons that were not pre-registered are marked post hoc.

J.2 The task, the labels and the public data

Task. A submission is a function that receives a model identifier, a reasoning effort and a list of problems, and returns one robustness verdict per problem; it runs offline in the organizers' container on a single GPU, with a time limit of one hour (one hour and ten minutes since 15 October, when evaluation moved to one call per problem) [4, 5, 6]. Five models are evaluated in the main phase: Qwen3.5-4B, Skywork-OR1-Math-7B, Olmo-3-7B-Think and DeepSeek-R1-0528-Qwen3-8B in both tracks, and gpt-oss-120b in the Main track only. Since 16 September the private validation tasks have 21 items (Main) and 14 (Small); the final test will use 30 to 50 new problems from unpublished AIMO sets [5, 6].

Labels. On every one of the 379 labeled rows of the earlier public datasets, the label equals the conjunction "no perturbation lowered accuracy" (no detrimental permutation; permuted accuracy 1.0). A majority rule fits these rows equally well, because none has permuted accuracy strictly between 0.5 and 1; the organizers' proposal defines a majority over perturbed variants [1], and the website states that the exact criterion differs from that outline and is not published [4]. The main-phase train set (16 September; 82 rows, 71 labeled in the revision we used, extended to 94 rows on 15 October) uses thresholds on the largest accuracy drop across perturbation families: robust at most 0.10, non-robust at least 0.25, unlabeled in between, also in the validation and test splits [5]. We use the conjunction for the earlier data and report the new rule where it applies (Section 5).

Public data. Six datasets aggregate to 1,196 rows over 335 (model, problem) pairs and 174 problems; 52 pairs repeat the DeepSeek measurements under a second model name, so 283 are distinct. For DeepSeek, the only evaluated model with substantial ground truth under the earlier rule, 164 pairs are labeled: 138 declared and 26 derived by us under the conjunction (a majority rule would call 21 of these 26 robust rather than 4); every analysis on these labels inherits that choice. Ground truth for the earlier rule exists otherwise only for 10 gpt-oss-120b pairs at low reasoning effort, five of them with a declared label. Two properties of the public data matter below. In the three aggregated datasets, the field `permutation_type` is a JSON list if and only if the item is robust, because the aggregation merges all perturbation types for robust items and keeps only the type or types with the lowest perturbed accuracy otherwise: a classifier using it is near-perfect in development and has no such field at inference (the organizers dropped it from their import script on 1 September; it remains in the datasets). And the public validation sample, `val-sample`, contains no labeled pair with base accuracy below 0.05, while 20 of the 80 robust pairs among the 335 published pairs, all of them in `aimo-interp-challenge-sample-full`, have base accuracy 0 and are robust only trivially under the conjunction.

Floors. On 21 and 14 items the binomial standard error of an accuracy near one half is 10.9 and 13.4 points. On the validation tasks introduced on 16 September the robust base rate is 10/21 and 7/14 (Section 6), so a constant scores 11/21 and 7/14 at best. A difference of one item is 4.8 and 7.1 points.

J.3 The falsification campaign

Validation that fits on some problems and tests on others is grouped by problem. Table J.1 condenses the campaign; Appendix A gives every entry with its baseline and caveats.

Table J.1. Channels tested against ground truth (DeepSeek and the public pairs unless stated). Accuracy differences are in points against the baseline named in each pre-registration. H1, H7 and T are computed on the 335 published pairs, counting the 52 duplicated DeepSeek pairs twice; counted once, H1 is −7.4, H7 −1.4 and T +7.1, and no verdict changes.

Channel	Entry	Result	Verdict
Problem text, 32 features, grouped 5-fold	H1	0.675 vs 0.761 (−8.7)	dead
Robustness as a property of the problem	H2	−1.3 (z = 0.88) with duplicated rows counted once; +3.4 (z = 4.04) only when counted twice	dead
Labels, base accuracy, difficulty order across models	H3 to H5, H7	−7.3, −11.8, −25.4, −1.2	dead
Official uncertainty features, logistic regression, grouped CV	H10	0.7744 vs 0.756, inside one standard error	reaches the proposal's accuracy (method substituted)
Cached features for other models; per-target thresholds	H12, H13	−16.5; +7.3 from thresholds chosen on the scored items, −2.8 when forced to rank	dead; ceiling withdrawn
Our families; translation; self-consistency; the organizers' families regenerated (pilot)	H19, H22, H33, H32	kappa 0.02; kappa −0.10; AUROC 0.502; kappa −0.001	dead
Representation shift under perturbation; length control	H30, H31	AUROC 0.615 (planned threshold 0.655; character count 0.705); 0.555	dead
Our linear probe on the residual stream, nested CV	H36	AUROC 0.710; about +0.02 over character count	survives, largely explained
The organizers' published probe artifact	H35	AUROC 0.739 (pre-registered aggregation) and 0.709 (its own), 0.62 and 0.57 residualized on log character count; shuffled-label controls 0.68	survives, largely explained
Trivial-robust rule (robust iff base accuracy < 0.05), post hoc	T	+6.0 on the 335 public pairs, +0.0 on `val-sample`	trap
Identical bundles re-executed (warmup, 17 items)	H18	11, 9, 11 and 11 correct	noise confirmed

Three traps a conventional validation would not catch. Rule T gains six points on the public pairs and nothing on `val-sample`, because the trivially robust subpopulation is absent there (Section 2). The oracle H13 appeared to show cross-model signal (+7.3 points, still +5.5 when constant predictions are excluded), but each target's threshold was chosen among all cut points on the very items it was scored on; forced to rank the items and to predict the true number of positives, it gave −2.8. And H2, the only warmup hypothesis about a property of the problem that had survived, owed its survival to one measurement published under two model names and counted twice: counted once, its concordance is no longer significant against a null with per-model robust rates (z = 0.88), and predicting from it scores 1.3 points below the per-model majority.

Length. Problem length recurs throughout: the character count alone orders the DeepSeek labels at AUROC 0.705, above the representation-shift statistic of H30 (0.615). Our pre-registered entry V8 (Section 6) is the rule `len(problem) >= 98`, with the threshold at the lower median of the 174 public problems and the direction fixed on the public pairs (balanced accuracy 0.652 over the 335 published pairs, 0.631 over the 283 distinct ones).

Probes. Our logistic probe on last-token hidden states of DeepSeek, with layer and regularization chosen in an inner cross-validation, reaches AUROC 0.710 out of fold and survives its permutation null (p95 0.574, with 60 permutations where 500 were pre-registered; 0.570 with the full regularization grid). The organizers' artifact survives its null as well (Table J.1), although our pre-registration predicted that both probes would die. The length controls that follow, like our analyses of the artifact (its own inference aggregation, its control groups and the directions of its probes), were not pre-registered: we added them after seeing the data. The probe's score correlates with problem length at Spearman +0.55; in a second-level cross-validation over 20 splits, character

count alone gives 0.696, the probe 0.697 and the two together 0.714, a gain of about two points; inside the tercile of the longest prompts the probe is at 0.537, indistinguishable from chance. What the probe reads is not established: residualized on log character count its score gives 0.62, and log character count residualized on the probe gives 0.62 as well. The organizers' published DeepSeek probe artifact orders our labels at AUROC 0.709 under its own inference aggregation, but 0.57 once residualized on log character count, and its own randomization controls, trained on shuffled labels, give 0.68 by themselves; at the layer it uses, each seed's probe points the same way whether trained on true or shuffled labels (cosine 0.93 to 0.998). Under its own inference aggregation, then, the artifact orders the labels mostly through problem length; under the aggregation we pre-registered (AUROC 0.739) its residual on log character count is 0.62, as for our probe, which is largely redundant with length as a predictor of the label.

J.4 Re-execution noise on the evaluation workers' GPU model

On 8 September the organizers provided a virtual machine with two NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs (96 GB each), the same model as the evaluation workers. All runs in this section used the official runtime image (torch 2.12.1, transformers 5.13.0) and the unmodified official extraction and regressor, on DeepSeek.

Table J.2. Forty public problems, twenty-three executions in six configurations; generated text hashed per item. Items crossing: items whose prediction changes across the runs of a configuration (crossings grow with the number of runs; per pair of runs, 15.3 under contention, 13.3 with batch size 1, 12.7 on the second card, 11.4 by default). Deterministic algorithms: `CUBLAS_WORKSPACE_CONFIG=:4096:8`, `torch.use_deterministic_algorithms(True, warn_only=True)` and deterministic cuDNN; compilation disabled: `TORCH_COMPILE_DISABLE=1`.

Configuration	Runs	Byte-identical items	Items crossing	Seconds per item
Default (batch 8)	5	0/40	24/40	26.0
Batch size 1	3	1/40	20/40	72.3
Two concurrent processes on one card	6	0/40	32/40	54.7
Second physical card	3	0/40	19/40	26.0
Deterministic algorithms and cuBLAS workspace setting	3	40/40	0	27.4
Compilation disabled	3	40/40	0	27.8

Crossings can only grow with the number of runs, so rows with different run counts are not directly comparable; averaged over pairs of runs, 15.3 items cross under contention, 13.3 with batch size 1, 12.7 on the second card and 11.4 under the default configuration.

Fourteen states. The twenty-three executions fall into fourteen distinct numerical states. Within a state all forty generations are byte-identical; different states share few: 61 of the 91 pairs of states share no generation and 28 share one or two, while one contention pair shares 9 and two batch-1 runs share 38. The state belongs to the process, not to the card: two runs launched in the same second on the same card landed in different states, while two of three runs on the second card reproduced the first card's default state exactly. We infer, without proof, that the state is fixed when the process starts, by a choice of kernels or algorithms that we did not identify. Two pre-registered predictions failed. Batch size 1 did not make generation more repeatable: its three runs gave three different states, with 1 of 40 items byte-identical across all three, while three of the five default runs were byte-identical to one another. The configuration without compilation, added during the run, was predicted to be byte-identical across its runs and to land in one of the default states; it was byte-identical, but in a state of its own. Because only 7 of the 40 problems are robust and crossings in both directions largely cancel, accuracy moved much less than predictions (from 21 to 24 correct under the default configuration on either card, and from 24 to 28 under contention), always below the 33 of the majority constant.

Candidate remedies. In three sequential runs on one card, either deterministic algorithms with a fixed cuBLAS workspace or disabling compilation made generation byte-identical at 5 to 7% runtime cost. Neither was tested under contention or across cards, and three identical sequential runs is also

what the default configuration gave before its fourth run diverged, so these are candidates, not fixes. Determinism also does not buy correctness: the two deterministic configurations settled on two different states, predicting 15 and 12 of 40 problems robust, with no generation in common and the same prediction on only 21 of 40. They also ran on different cards (deterministic algorithms on the first, compilation disabled on the second), so configuration and card are not separated, although the default state did reproduce across cards. The published number under any deterministic setting depends on which setting was chosen.

The official score channel. The official features and regressor score were extracted three times for the 174 public problems with the same code and image (this measurement was not pre-registered and is exploratory). Two runs landed in the same state; the third, flag-free like the second, matched it on 6 of 174 scores. The scores of the two flag-free runs correlate at Spearman 0.24 over the 164 labeled problems, and in the orientation of the regressor's own decision rule the extractions give AUROC 0.487 and 0.602. On 164 problems the difference is not significant (paired bootstrap 95% interval -0.015 to 0.246): the channel's ordering does not reproduce between identical runs, which is not the same as a demonstrated change of signal. Calibration moves with the state too: the shipped threshold predicts 25.6% of problems robust in one state and 33.5% in the other, against a base rate of 24.4%. Since 137 of the 164 problems are in the regressor's own training set, this is a statement about the stability of its inputs, not a test on new problems.

J.5 Can the label be reproduced?

If the label could be regenerated from the rule its public labels follow, it could be learned for the three evaluated models that had no public labels before the main-phase train set. We identified the organizers' five perturbation families from the public data (rephrase, rename, domain, distract, typos), regenerated them with gpt-oss-120b at low reasoning effort prompted with the public examples, and gated each variant with gpt-oss-120b at medium effort (the text must differ, the length ratio must lie between 0.7 and 4.0 and the solver's answer must not change; the prompts are in Appendix I). Because the public datasets carry no answers, an answer counts as correct when it matches gpt-oss-120b's answer on the original problem. On the organizers' virtual machine (E13), we labeled four models with ten variants per family in two independent draws (A and B; about 40 accepted variants per problem), ten samples on the base problem and one per variant, and re-sampled draw A's variants with a new seed (A′). Generation, gating and labeling ran on that machine in our own Python environment with vLLM 0.28.0 (torch 2.13.0, transformers 5.16.1, as recorded on 17 September), outside the official runtime image; variants were generated at temperature 1.0 with at most 4,096 tokens and gated at temperature 1.0 with at most 8,192 tokens.

Table J.3. Agreement of regenerated labels (Cohen's kappa; in brackets, 95% percentile bootstrap intervals over problems, 2,000 resamples). Robust rates are over all 164 problems (the 4 without accepted variants count as not robust); kappas are over the problems labeled in both series (159 for A and B, 160 for A′ and for the official labels); the main-phase column uses the problems labeled in both draws under the 0.10/0.25 thresholds.

Model	Robust in draw A	kappa(A, B), conjunction	kappa(A, A′), same variants, new seed	kappa(A, B), main-phase rule	kappa with official labels
Skywork	72.6%	0.439 [0.28, 0.59]	0.789 [0.67, 0.89]	0.49 [0.17, 0.76] (132 problems)	no earlier labels
DeepSeek	37.8%	0.413 [0.26, 0.55]	0.464* [0.33, 0.59]	0.56 [0.36, 0.73] (100)	0.025 [−0.12, 0.18]; 0.02 [−0.09, 0.13] under the main-phase rule (133)
Olmo	65.9%	0.378 [0.21, 0.54]	0.653 [0.52, 0.78]	0.41 [0.09, 0.69] (133)	no earlier labels
Qwen3.5-4B	9.1%	0.178 [−0.06, 0.39]	not run	0.34 [0.13, 0.53] (86)	no earlier labels

(*) On DeepSeek, A′ also received the answer-format line of Section 5.3, so this entry mixes seed and prompt.

5.1 More variants do not help, and our noise does not explain the gap. Subsampling DeepSeek's variants from 2 to 40 per problem (20 draws per size), the mean kappa(A, B) stays between 0.36 and 0.47 with no upward trend, and the mean concordance with the official label

between −0.02 and +0.03. Correcting the observed 0.025 for our process's unreliability gives 0.039; even at the upper end of its interval the corrected value is about 0.27, so our noise could hide at most a weak agreement. Under the main-phase rule (E13b, pre-registered on the same answers with no new generation), all three hypotheses died and none of our three predictions held in full: test-retest stays below 0.60 on every model, and, post hoc, the change of rule adds only 0.01 to 0.11 when both rules are read on the same problems. The gap is not only one of severity: under the main-phase rule, on the compared problems our DeepSeek labels are robust on 71 to 75% against the organizers' 28 to 30%, but, in comparisons made post hoc, our robust rate is about the same whether their label is robust or not (77.5% and 74.2% in draw A), and the largest accuracy drops do not correlate (Spearman −0.09 and −0.01), so no stricter threshold on our side would recover theirs.

5.2 What drives the instability. On Skywork, re-sampling the same variants with a new seed leaves much higher agreement than drawing new variants (0.79 against 0.44; under the main-phase rule and post hoc, 0.94 against 0.47 on the problems labeled in all three series), so the label depends mainly on which variants are drawn. On Olmo the seed alone accounts for about half of the instability under either rule, although under the main-phase rule this rests on 5 and 9 disagreements, too few to separate the two sources. A plug-in simulation of the organizers' process from their published prediction counts gives a test-retest kappa of 0.94 (median 0.93), but it resamples answers to the same variants and never lets a robust item fail; run on our own draw A, the same procedure gives 0.64 to 0.84, against a measured 0.18 to 0.44 under new variants. It overstates by about 0.4 on every model, so the reliability of the official label remains unmeasured. The files behind the official DeepSeek labels also used a different sampling regime: on 154 of the 164 problems most variants were answered ten times each and the base between 1 and 60 times, where we answered each variant once, which by itself makes a conjunction label stricter; and our decoding (temperature 0.6, top_p 0.95, at most 16,384 tokens) differs from theirs (temperature 1.0, up to 100,000 tokens), published after our runs [4]. What the data show is narrower than a verdict on the label: variants produced by our process measure something else. Whether any process reproduces the label is open; on 22 September the organizers wrote that they have decided not to publish the code that reproduces their labeling, to test the generalization of submitted methods [6].

5.3 An unspecified prompt moves the label. Fed the bare problem text, Qwen3.5-4B reasons for about eight thousand tokens and closes its answer in the expected format in only 5 of its 1,640 base samples in draw A. With one added instruction line requesting the format, the share of problems passing the base condition of the label (at least 6 of 10 samples agreeing with the reference answer) moved from 0 of 164 to 145 of 164 (88%), and its robust rate from 0% to 9.1%; DeepSeek passes the base condition on 147 of 164 problems with or without that line. A label defined by correctness is not comparable across prompts that the benchmark leaves unspecified.

J.6 Our entries, and the evaluations the organizers requested

v6, the measuring entry. It loads each evaluated model, extracts the official features from the model's own greedy generation and applies the official regressor (with a gpt-oss threshold fitted at run time on the ten gpt-oss rows of `val-sample`, whose only robust label is on one of three rows of `huikang-gpt-oss-120b-aimo3:low`, by its name a fine-tune of gpt-oss-120b). On the original main-phase tasks it scored 11/21 and 10/19, with runtimes of 838 s and 509 s.

V8, the entry that does nothing. Pre-registered on 13 September with conditional predictions, its entire logic is `return [len(p) >= 98 for p in problems]`: no model is loaded and each prediction depends only on its own item. On the original tasks it scored 11/21 and 10/19, identical to v6 and to the organizers' own entries on the leaderboard (0.52 and 0.53); on 21 and 19 items these ties are compatible with length carrying little signal (95% Wilson intervals 0.32 to 0.72 and 0.32 to 0.73) without showing that it carries none, and none of these entries scored above the majority constant (at least 11/21 and 10/19). A one-line rule that ties a pipeline loading up to five models for 8.5 to 14 minutes is a measurement of the pipeline. On the replacement tasks of 16 September it predicted robust for every item and scored 10/21 and 7/14, the robust base rates.

The evaluations requested for reports [5]. On the public main-phase train set (the revision of 17 September), V8 predicts robust for 65 of 71 labeled rows and scores 46/71 (0.648), below the 52/71 (0.732) of the all-robust constant (balanced accuracy 0.442). On the private validation set it scored

10/21 and 7/14 under the original batch protocol (21 September). On 15 October the organizers moved validation to one call per problem, erased the leaderboard and re-evaluated a portion of the top submissions [6]; on 28 October we resubmitted V8 unchanged (Codabench submissions 1016713, Main, and 1016730, Small), with an all-robust constant as control (1016732 and 1016735). Called once per problem through its legacy signature `are_robust(model_id, problems)`, V8 again predicted robust for every item, as its logged counts show, and scored 10/21 and 7/14, the robust base rates; the constant scored the same. For the final test we nominated V8 (1016713) and the all-robust constant (1016732), both on the Main track; their scores were not yet available when we submitted this report. Our pre-registered predictions for the final evaluation (Appendix B) will be scored there: P1, that the top-ranked entry's final accuracy lies within two binomial standard errors of the test set's majority constant, and P2, about the entries that led on 7 September. If P1 or P2 fail, the label has learnable structure that we did not find.

J.7 Recommendations

R1. Fix the numerical state and test the fix under load. Either configuration of Section 4 made the pipeline byte-reproducible in sequential runs at 5 to 7% cost; it should be verified under the workers' real load and across cards before it is relied on, and the chosen setting documented, since it selects the fixed point. Isolating submissions on a card may matter as much: two processes on one card were the worst case measured.

R2. Repeat only where it decides something. With the re-execution standard deviation measured in the warmup (1.0 item on 17, from four runs), resolving a one-item gap needs about eight runs, though a standard deviation estimated from four runs is itself uncertain, and so is that number; repeating only the entries near a prize boundary at the close costs a handful of runs rather than three times the field.

R3. Publish the labeling prompt, at least after the close. For Qwen3.5-4B the answer-format instruction moves the share of problems passing the base condition from 0% to 88% (Section 5.3).

R4. Measure the label's reliability directly. Two independent draws of variants, labeled with the official process, would give the test-retest agreement that the plug-in simulation cannot (Section 5.2); until then, the reliability of the official label remains unmeasured.

R5. Show the floors and the noise on the leaderboard. A row per constant predictor, a stated re-execution standard deviation and binomial intervals; on 21 and 14 items the standard error is 10.9 and 13.4 points.

R6. Require a length control in white-box ablations. The White-Box Awards announced on 16 October ask for an ablation showing that the white-box signal matters [6]. On DeepSeek, the organizers' artifact, under its own inference aggregation, ordered the labels mostly through problem length, and our own probe was largely redundant with length as a predictor, adding about two points of AUROC to the character count, with what it reads not established (Section 3); an ablation that removes the white-box component without controlling for length cannot separate the white-box signal from length.

R7. Remove `permutation_type` from the aggregated public datasets (Section 2).

K Codabench submissions, bundles and release

Submissions. Every submission under our Codabench account (user `dan23rr`) in both phases, read from the Codabench API on 28 October 2026. Accuracy is given as items correct over items; a parent submission sends one bundle to both tasks, with one child per task it was evaluated on (the parents of 13 September also carry the cancelled children listed below the table).

ID	Task	Date (UTC)	Accuracy	What it is
907310	warmup	2026-08-29	8/17	Null probe, every item non-robust; floor pre-registered at 8/17 (full report, Section 3.1).
907677	warmup	2026-08-29	9/17	Official baseline, unchanged, instrumented to log its runtime (Appendix D).
907967	warmup	2026-08-29	9/17	Early version of the measuring entry, with a time budget, as its released bundle documents; its logs give the container's GPU and library versions (Appendix D).
909567	warmup	2026-08-31	9/17	Later version of the same entry, as its released bundle documents.
909843	warmup	2026-08-31	9/17	V4, the allocation entry, submitted once as a measurement against its pre-registered rule (Appendix G).
909853	warmup	2026-08-31	10/17	v5: every model measured by itself (H16, Appendix A).
909976	warmup	2026-08-31	11/17	v6: H16 with a gpt-oss threshold calibrated on `val-sample` (H17); with the next three, the four executions of H18.
909994	warmup	2026-08-31	9/17	v6, re-executed (H17, H18).
910002	warmup	2026-08-31	11/17	v6, re-executed (H17, H18).
910013	warmup	2026-08-31	11/17	v6, re-executed (H17, H18).
912096	Main (original)	2026-09-02	11/21	v6, the measuring entry, on the original Main task (Section 6).
912097	Small (original)	2026-09-02	10/19	v6 on the original Small task (Section 6).
924968	both tasks (parent)	2026-09-13	parent	V8 Main bundle, sent to both original tasks: parent of 924969 and 924970.
924969	Main (original)	2026-09-13	11/21	V8 on the original Main task (Section 6).
924970	Small (original)	2026-09-13	coverage 0	V8 Main bundle on the Small task, rejected before import because it lacks `small.txt` (coverage 0).
924972	both tasks (parent)	2026-09-13	parent	V8 Small bundle, sent to both original tasks: parent of 924973 and 924974.
924973	Main (original)	2026-09-13	coverage 0	V8 Small bundle on the Main task, rejected before import because of `small.txt` (coverage 0).
924974	Small (original)	2026-09-13	10/19	V8 on the original Small task (Section 6).
925822	Main (original)	2026-09-14	11/21	V8 Main bundle again, Main task only, uploaded by the AI assistant without a specific instruction (AI usage statement).
936802	Main	2026-09-21	10/21	V8 on the replacement Main task (Section 6, Appendix B).
936806	Small	2026-09-21	7/14	V8 on the replacement Small task (Section 6, Appendix B).
1016713	Main	2026-10-28	10/21	V8, unchanged, on the current validation task, after the change of 15 October (Section 6); nominated for the final test, Main track.
1016730	Small	2026-10-28	7/14	V8, unchanged, on the current validation task, after the change of 15 October (Section 6).
1016732	Main	2026-10-28	10/21	All-robust constant, control (Section 6); nominated for the final test, Main track.
1016733	both tasks (parent)	2026-10-28	parent	Constant Small bundle: parent of 1016734 and 1016735.
1016734	Main	2026-10-28	coverage 0	Constant Small bundle sent to the Main task in error (AI usage statement); rejected before import ("small.txt is not allowed for a main-track submission"), as 924973 was on the original Main task.
1016735	Small	2026-10-28	7/14	All-robust constant, control (Section 6).

Also under our account, and not described in this report: 928172, 928173, 928174, 928175, 929013 and 929014, children of 924968 and 924972 created on 16 September and cancelled, without scores.

Bundles. Our entry on the current tasks is V8, with the all-robust constant as control; we nominated 1016713 and 1016732 for the final test, on the Main track. Neither loads a model or reads data at run time. V8's two parameters were fixed on the six public datasets, which are its training data; Appendix D (full report, Section 13) gives them, the scripts that recompute them and the instructions to reproduce its score. SHA-256 of each bundle:

- `solution-v8-main.zip`: V8: a single `solution.py` defining `are_robust(model_id, problems)`, which returns `[len(p) >= 98 for p in problems]`. Select only the Main task.
  `7b70f15c51752dcf4285e477d58c541d0ecf052d833c325c0702dbbd054ffb5f`
- `solution-v8-small.zip`: The same `solution.py`, with the empty marker file `small.txt` that the Small task requires. Select only the Small task.
  `0b83b55f94dc16b9852a7c2b35d71ac5787e43d81afd6396dee00fb5e011e500`
- `solution-costante-main.zip`: The all-robust constant: a single `solution.py` whose `are_robust` returns `True` for every problem, written to accept the argument orders of both official descriptions of the interface and the legacy two-argument call. Select only the Main task.
  `35c47bbef717a5d4c5b6c7786bddbab9d87a02483e4f417cd34fd4ecf669c67d`
- `solution-costante-small.zip`: The same `solution.py`, with `small.txt`. Select only the Small task.
  `f798c2979b136a84c22e6d994bf0228bbfeffa044ac68f7868e13859b4c6075a`

The V8 docstring, frozen with the bundle on 13 September, contains three statements that this report corrects: it calls 98 the median of the character counts (it is the lower median) and gives the rule's balanced accuracy as 0.658, the figure for a threshold of 99 (Appendix C, item 37); and it says that the official score changes sign between identical runs, quoting AUROC 0.513 and 0.398 with robust as the positive class on the raw score, where Section 4 gives the same runs as 0.487 and 0.602 in the regressor's own orientation and finds the difference not significant.

Reproducing our score. Upload each bundle with only its own task selected: the Main bundle on task 36237, the Small bundle on task 36241. No model and no data are needed. On the current validation tasks the expected results are those of 1016713 and 1016730, 10/21 and 7/14; Appendix D gives the instructions and expected results of 23 September, before the change of 15 October.

Release. Every pre-registration, outcome file and analysis script, the extraction outputs, the submission bundles as submitted, our perturbed variants of the public problems, our snapshots of the Codabench leaderboard and the full report (draft v2.4) are released with this report at https://github.com/Dan23RR/aimo-instrument, which keeps the layout of our working folder so that the file paths cited here are valid there, with other participants' names, usernames and passages about their entries redacted. The organizers' getting-started files inside our bundles and our variants are released with their permission (Appendix H); copies of the organizers' datasets are not, and in our variant files the rows of the original problems keep their identifiers but not their texts, while five original problems remain, as submitted, in the calibration file of the v6 bundles. Our correspondence, internal notes and the drafts before v2.4 that Appendix D lists are not released. On 7 September we asked the organizers whether measurements taken on their virtual machine may appear in this report and undertook to share our results with them, and we repeated the question on 2 November; the same day they answered that this is acceptable, and the results reach them with this report and the release. Deviations from the pre-registrations are listed in Appendix D.

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regularization by inner cross-validation (Section 3), the decoding, sampling counts and prompts of the regeneration (Section 5, Appendices I and J) and the runtime configurations (Section 4, Table 2) are given.

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invalidated Qwen runs (about fifteen card-hours), while the totals for E14 to E16 and the billed totals are still to be itemized from the run logs and billing records.

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    Justification: Language models are the objects of study (Section 2); in our label regeneration gpt-oss-120b generated and gated the perturbation variants and supplied the reference answers (Section 5, Appendix I), and phi-4 translated problems for H22 (Appendix H). An AI assistant (Claude, through Claude Code) drafted the pre-registrations, including hypotheses, predictions and kill criteria, wrote and ran the code, operated the virtual machine, made most of our uploads and ran the adversarial reviews behind most corrections, as disclosed in the AI usage statement.

    Guidelines:

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